Full Conspect in Mathematical Analysis

A generalized, topologically formal, proof-complete version

Riemann–Darboux Integration, Paths, Improper Integrals, Series, and Products

The purpose of this handout is not to list formulas. Each ticket is written as a small lecture: definitions, structural context, theorem statements, complete proofs, and practice problems. The underlying language is intentionally general: order completeness of the real line, compactness of intervals, directed sets of partitions, Cauchy criteria, normed linear spaces, bounded linear functionals, uniform convergence, and Abel summation.

Generated: June 16, 2026

Contents

Foundational layer 2

1 Primitives and the indefinite integral 2

2 Partitions, tagged partitions, Riemann sums, and boundedness 4

3 Darboux sums and Darboux integrals 5

4 Darboux criteria for integrability and monotone functions 7

5 Oscillation and the oscillation criterion 8

6 Cantor theorem and integrability of continuous functions 9

7 Algebra of integrable functions 10

8 Monotonicity of the integral, absolute value, and finite changes 12

9 Additivity over intervals and the mean value theorem 13

10 Variable upper limit and the Newton–Leibniz formula 14

11 Definite integration by parts, substitution, and Taylor's formula 15

12 Paths in $\mathbb{R}^d$: concatenation, equivalence, and length 16

13 Length of a smooth path 17

14 Improper integrals and the Cauchy criterion 18

15 Absolute convergence and comparison tests for improper integrals 19

16 Dirichlet and Abel tests for improper integrals 20

17 Numerical series: Cauchy criterion, geometric series, and basic properties 21

18 Absolute convergence and root/ratio tests for series 22

19 Integral test, Dirichlet test, and Abel test for series 23

20 Rearrangement of absolutely convergent series 24

21 Products of absolutely convergent series 25

Final synthesis 26

Appendix A. Detailed proof bank and structural expansions 26

Foundational layer

Core idea. The whole course is controlled by a small number of structural principles: completeness of $\mathbb{R}$, compactness of closed intervals, Cauchy criteria, and continuity of linear operations in normed spaces.

Order and compactness of intervals

We use the least-upper-bound property of $\mathbb{R}$: every nonempty subset of $\mathbb{R}$ bounded above has a supremum. Consequences used repeatedly are: monotone convergence of bounded monotone sequences, Bolzano–Weierstrass, Heine–Borel compactness of $[a, b]$, and Cantor's theorem that continuous functions on compact metric spaces are uniformly continuous.

Directed sets and filters of partitions

A directed set is a preorder $(D, \leq)$ such that any two elements have a common upper bound. A net $(x_d)_{d\in D}$ in a topological space converges to x if for every neighborhood U of x, eventually $x_d \in U$. For Riemann integration, the directed objects are tagged partitions ordered by refinement or by decreasing mesh. Thus the statement

$$\lim_{\text{mesh}\, P\to 0} S(f, P, \xi) = I$$

means: for every $\varepsilon > 0$ there is $\delta > 0$ such that every tagged partition of mesh less than δ has Riemann sum within ε of I. This is a filter limit over the set of all sufficiently fine tagged partitions.

Normed-space language

The space $B(E)$ of bounded scalar functions on a set E, with norm

$$\|f\|_\infty = \sup_{x\in E} |f(x)|,$$

is a normed vector space. Uniform convergence of functions is convergence in this norm. On $C([a, b])$, the integral

$$I(f) = \int_a^b f(x)\, dx$$

is a bounded linear functional, and many theorems about limits under integrals become one sentence: continuous linear maps preserve limits.

Ticket 1. Primitives and the indefinite integral

Core idea. A primitive is a global inverse operation to differentiation on an interval. The reason indefinite integrals form a family with an arbitrary constant is the mean value theorem.

Definition 1.1 (Primitive)**.** *Let $I \subset \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. A function $F : I \to \mathbb{R}$ is a primitive, or antiderivative, of f on I if F is differentiable on I and*

$$F'(x) = f(x) \qquad (x \in I).$$

The notation

$$\int f(x)\, dx = F(x) + C$$

means the family of all primitives of f on the interval.

Theorem 1.1 (Description of all primitives)**.** *Let I be an interval. If F and G are primitives of f on I, then there exists a constant $C \in \mathbb{R}$ such that*

$$F(x) = G(x) + C \qquad (x \in I).$$

Conversely, if F is a primitive of f, then $F + C$ is also a primitive for every constant C.

Proof. Let $H = F - G$. Then $H'(x) = F'(x) - G'(x) = f(x) - f(x) = 0$ for all $x \in I$. If $x < y$ are points of I, the mean value theorem applied to H on $[x, y] \subset I$ gives a point $c \in (x, y)$ such that

$$H(y) - H(x) = H'(c)(y - x) = 0.$$

Thus $H(y) = H(x)$ for all $x, y \in I$, so H is constant. The converse is immediate because $(F + C)' = F' = f$. ☐

Theorem 1.2 (Linearity of the indefinite integral)**.** *If $F' = f$ and $G' = g$ on an interval I, then for scalars α, β,*

$$(\alpha F + \beta G)' = \alpha f + \beta g.$$

Hence

$$\int (\alpha f + \beta g)\, dx = \alpha \int f\, dx + \beta \int g\, dx.$$

Proof. This is exactly linearity of the derivative:

$$(\alpha F + \beta G)' = \alpha F' + \beta G' = \alpha f + \beta g.$$

The equality of indefinite integrals is equality of families of primitives, up to an arbitrary additive constant. ☐

Theorem 1.3 (Substitution for primitives)**.** *Let $F' = f$ on an interval J, and let $\varphi : I \to J$ be differentiable. Then*

$$(F \circ \varphi)'(t) = f(\varphi(t))\varphi'(t),$$

so

$$\int f(\varphi(t))\varphi'(t)\, dt = F(\varphi(t)) + C.$$

Proof. This is the chain rule:

$$(F \circ \varphi)'(t) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t).$$

Therefore $F \circ \varphi$ is a primitive of the integrand. ☐

Theorem 1.4 (Integration by parts for primitives)**.** *If u, v are differentiable on I, then*

$$\int u'(x)v(x)\, dx = u(x)v(x) - \int u(x)v'(x)\, dx.$$

Proof. The product rule gives

$$(uv)' = u'v + uv'.$$

Thus $u'v = (uv)' - uv'$. Taking primitives gives the formula. ☐

Generalized viewpoint. Ticket 1 is a statement about the kernel of the derivative operator. On an interval, $D(F) = F'$ has kernel equal to the one-dimensional space of constant functions. Therefore primitives, if they exist, form an affine space $F + \ker D$.

Practice problems.

1. Prove directly, using the mean value theorem, that if $H' = 0$ on an interval, then H is constant. Explain why the interval assumption is essential.
2. Find all primitives of xe^{x^2} by substitution and justify every step as a statement about derivatives.
3. Derive the integration-by-parts formula for $\int uv'\,dx$ from the product rule and compare it with the formula above.

Ticket 2. Partitions, tagged partitions, Riemann sums, and boundedness

Core idea. Riemann integration is convergence of a net of finite weighted sums as the partition becomes finer.

Definition 2.1 (Partition, mesh, tagged partition)**.** *A partition of $[a, b]$ is a finite ordered set*

$$P : a = x_0 < x_1 < \cdots < x_n = b.$$

Its mesh is

$$\mathrm{mesh}(P) = \max_{1\le i\le n}(x_i - x_{i-1}).$$

A tagged partition is a pair (P, ξ), where $\xi_i \in [x_{i-1}, x_i]$. For a function $f : [a, b] \to \mathbb{R}$, the Riemann sum is

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1}).$$

Definition 2.2 (Riemann integrability by fine tagged sums)**.** *A function $f : [a, b] \to \mathbb{R}$ is Riemann integrable with integral I if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every tagged partition (P, ξ),*

$$\mathrm{mesh}(P) < \delta \quad \Longrightarrow \quad |S(f, P, \xi) - I| < \varepsilon.$$

We then write

$$I = \int_a^b f(x)\,dx.$$

Theorem 2.1 (Uniqueness of the integral)**.** *If a Riemann integral exists, it is unique.*

Proof. Suppose I and J both satisfy the definition. Given $\varepsilon > 0$, choose δ_I and δ_J corresponding to $\varepsilon/2$. Choose any tagged partition with mesh less than $\min(\delta_I, \delta_J)$. Then

$$|I - J| \le |I - S(f, P, \xi)| + |S(f, P, \xi) - J| < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, $I = J$. □

Theorem 2.2 (Cauchy criterion for Riemann sums)**.** *A function f is Riemann integrable if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that whenever (P, ξ) and (Q, η) are tagged partitions with mesh less than δ,*

$$|S(f, P, \xi) - S(f, Q, \eta)| < \varepsilon.$$

Proof. If f is integrable with integral I, choose δ so that every sufficiently fine Riemann sum is within $\varepsilon/2$ of I. Then two such sums differ by less than ε.

Conversely, assume the Cauchy condition. For each n, choose an arbitrary tagged partition (P_n, ξ^n) with $\mathrm{mesh}(P_n) < 1/n$. The Cauchy condition implies that the sequence $S(f, P_n, \xi^n)$ is Cauchy in $\mathbb{R}$, hence converges to some I. We prove that all sufficiently fine Riemann sums are close to I. Given $\varepsilon > 0$, choose δ from the Cauchy condition with $\varepsilon/2$, and choose n so large that $1/n < \delta$ and $|S(f, P_n, \xi^n) - I| < \varepsilon/2$. If (P, ξ) has mesh less than δ, then

$$|S(f, P, \xi) - I| \le |S(f, P, \xi) - S(f, P_n, \xi^n)| + |S(f, P_n, \xi^n) - I| < \varepsilon.$$

Thus f is integrable with integral I. □

Theorem 2.3 (Riemann integrable functions are bounded)**.** *If $f : [a, b] \to \mathbb{R}$ is Riemann integrable in the tagged-sum sense, then f is bounded.*

Proof. Assume f is unbounded. Fix $\varepsilon = 1$ in the Cauchy criterion and choose $\delta > 0$. Take any partition P with $\mathrm{mesh}(P) < \delta$. Since f is unbounded on $[a, b]$, at least one subinterval $[x_{j-1}, x_j]$ has unbounded f; otherwise finitely many bounded restrictions would make f bounded on the whole interval. Fix tags on all other subintervals. Because f is unbounded on $[x_{j-1}, x_j]$, choose two tags ξ_j, η_j in this subinterval such that

$$|f(\xi_j) - f(\eta_j)|(x_j - x_{j-1}) > 1.$$

The two tagged sums corresponding to the same partition and different tags on the j-th subinterval then differ by more than 1, contradicting the Cauchy criterion. Hence f must be bounded. □

Generalized viewpoint. The set of all tagged partitions ordered by refinement is a directed set. The Riemann integral is the limit of the net $S(f, P, \xi)$. The Cauchy criterion above says that this net is Cauchy in the complete space $\mathbb{R}$.

Practice problems.

1. Prove that the collection of partitions of $[a, b]$, ordered by refinement, is directed.
2. Using the Cauchy criterion, prove uniqueness of the Riemann integral without referring to a preselected sequence of partitions.
3. Give a direct example showing why tags matter in the definition of the Riemann sum.

Ticket 3. Darboux sums and Darboux integrals

Core idea. Darboux theory removes tags. It replaces arbitrary sample values by the best lower and upper estimates on each subinterval.

Definition 3.1 (Upper and lower Darboux sums)**.** *Let $f : [a, b] \to \mathbb{R}$ be bounded and let $P : a = x_0 < \cdots < x_n = b$. Define*

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x), \qquad M_i = \sup_{x \in [x_{i-1}, x_i]} f(x).$$

The lower and upper Darboux sums are

$$L(f, P) = \sum_{i=1}^{n} m_i (x_i - x_{i-1}), \qquad U(f, P) = \sum_{i=1}^{n} M_i (x_i - x_{i-1}).$$

Definition 3.2 (Lower and upper integrals). *For bounded f, define*

$$\underline{\int_a^b} f = \sup_P L(f,P), \qquad \overline{\int_a^b} f = \inf_P U(f,P),$$

where the supremum and infimum are taken over all partitions of $[a,b]$.

Lemma 3.1 (Refinement monotonicity). *If Q refines P, then*

$$L(f,P) \le L(f,Q) \le U(f,Q) \le U(f,P).$$

Proof. It is enough to split one subinterval $[u,v]$ of P at a point c. Let

$$m = \inf_{[u,v]} f, \quad m_1 = \inf_{[u,c]} f, \quad m_2 = \inf_{[c,v]} f.$$

Since $[u,c]$ and $[c,v]$ are subsets of $[u,v]$, we have $m \le m_1$ and $m \le m_2$. Hence

$$m(v-u) \le m_1(c-u) + m_2(v-c).$$

Thus lower sums do not decrease under this refinement. The upper-sum inequality is analogous, using suprema: $M_1, M_2 \le M$, so the upper contribution does not increase. Iterating over finitely many inserted points proves the claim. □

Lemma 3.2 (Lower sums never exceed upper sums). *For any two partitions P, Q,*

$$L(f,P) \le U(f,Q).$$

Proof. Let $R = P \cup Q$ be the common refinement. By refinement monotonicity,

$$L(f,P) \le L(f,R) \le U(f,R) \le U(f,Q).$$

□

Theorem 3.1 (Darboux convergence under small mesh). *Let f be bounded. Then as* $\mathrm{mesh}(P) \to 0$, *the upper sums tend to the upper Darboux integral and the lower sums tend to the lower Darboux integral:*

$$U(f,P) \to \overline{\int_a^b} f, \qquad L(f,P) \to \underline{\int_a^b} f.$$

Proof. We prove the statement for upper sums. Let $I^* = \overline{\int_a^b} f$ and choose a partition P_0 such that

$$U(f,P_0) < I^* + \varepsilon/2.$$

Let $|f| \le M$, and let P_0 have N interior points. For any partition P, form the common refinement $R = P \cup P_0$. Then $U(f,R) \le U(f,P_0) < I^* + \varepsilon/2$. The only difference between $U(f,P)$ and $U(f,R)$ occurs in subintervals of P that contain points of P_0. The total length of such affected intervals is at most $N\,\mathrm{mesh}(P)$. On each affected part, the possible change in upper contribution is bounded by $2M$ times length. Therefore

$$0 \le U(f,P) - U(f,R) \le 2MN\,\mathrm{mesh}(P).$$

Choose $\delta > 0$ such that $2MN\delta < \varepsilon/2$. If $\mathrm{mesh}(P) < \delta$, then

$$I^* \le U(f,P) \le U(f,R) + 2MN\,\mathrm{mesh}(P) < I^* + \varepsilon.$$

Thus $U(f,P) \to I^*$. The proof for lower sums is identical with inequalities reversed. □

Generalized viewpoint. Darboux sums form two monotone approximation procedures indexed by partitions. Refinement is the natural order; upper sums move downward and lower sums move upward. The upper and lower integrals are the corresponding extremal limits.

Practice problems.

1. Prove refinement monotonicity for a partition obtained by inserting exactly one point. Then explain why this proves the general case.
2. Prove that $\sup_P L(f,P) \le \inf_Q U(f,Q)$.
3. Fill in the detailed proof of the lower-sum analogue of the Darboux convergence theorem.

Ticket 4. Darboux criteria for integrability and monotone functions

Core idea. A bounded function is integrable exactly when its best lower and upper finite approximations can be forced arbitrarily close.

Definition 4.1 (Darboux integrability)**.** *A bounded function $f : [a,b] \to \mathbb{R}$ is Darboux integrable if*

$$\underline{\int_a^b} f = \overline{\int_a^b} f.$$

The common value is denoted $\int_a^b f$.

Theorem 4.1 (Equivalent Darboux criteria)**.** *For a bounded function $f : [a,b] \to \mathbb{R}$, the following are equivalent:*

1. *f is Darboux integrable.*
2. *For every $\varepsilon > 0$, there exists a partition P such that*

$$U(f,P) - L(f,P) < \varepsilon.$$

3. *The fine-mesh Riemann sums converge to one common number independent of tags.*

Proof. (1) $\Rightarrow$ (2). Let I be the common upper and lower integral. Choose partitions P_1, P_2 such that

$$U(f,P_1) < I + \varepsilon/2, \qquad L(f,P_2) > I - \varepsilon/2.$$

Let $P = P_1 \cup P_2$. By refinement monotonicity,

$$U(f,P) \le U(f,P_1) < I + \varepsilon/2, \qquad L(f,P) \ge L(f,P_2) > I - \varepsilon/2.$$

Thus $U(f,P) - L(f,P) < \varepsilon$.

(2) $\Rightarrow$ (1). Since $L(f,P) \le \underline{\int} f \le \overline{\int} f \le U(f,P)$, condition (2) gives

$$0 \le \overline{\int} f - \underline{\int} f \le U(f,P) - L(f,P) < \varepsilon.$$

Letting $\varepsilon \to 0$, the upper and lower integrals are equal.

(2) $\Rightarrow$ (3). Let $|f| \le M$. Choose a partition P_0 with $U(f,P_0) - L(f,P_0) < \varepsilon/3$. By the Darboux convergence theorem, every sufficiently fine partition P has upper and lower sums within $\varepsilon/3$ of the corresponding sums over a common refinement with P_0. Therefore every sufficiently fine tagged sum lies

between numbers whose difference is less than ε. Thus all sufficiently fine tagged sums are mutually ε-close; by the Cauchy criterion, they converge to a unique limit.

(3) $\Rightarrow$ (2). If fine tagged sums converge independently of tags, then for sufficiently fine P, any two tagged sums over P differ by less than ε. Taking tags whose values approximate the infimum and supremum on each subinterval gives $U(f,P) - L(f,P) \le \varepsilon$, up to an arbitrarily small error. Hence a partition with $U - L < 2\varepsilon$ exists. $\square$

Theorem 4.2 (Monotone functions are integrable)**.** *Every monotone function $f : [a,b] \to \mathbb{R}$ is Riemann–Darboux integrable.*

Proof. Assume f is increasing; the decreasing case is analogous. For a partition $P : a = x_0 < \cdots < x_n = b$, the infimum and supremum on $[x_{i-1}, x_i]$ are $f(x_{i-1})$ and $f(x_i)$. Hence

$$U(f,P) - L(f,P) = \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))(x_i - x_{i-1}).$$

If $\operatorname{mesh}(P) < \delta$, then

$$U(f,P) - L(f,P) \le \delta \sum_{i=1}^{n} (f(x_i) - f(x_{i-1})) = \delta(f(b) - f(a)).$$

Given $\varepsilon > 0$, choose $\delta < \varepsilon/(f(b) - f(a))$ if $f(b) > f(a)$, otherwise the function is constant. Then $U - L < \varepsilon$. By the Darboux criterion, f is integrable. $\square$

Practice problems.

1. Prove all implications in the Darboux integrability criterion without using sequences.
2. Show that every step function on $[a,b]$ is integrable and compute its integral.
3. Prove the integrability of decreasing functions by reducing to the increasing case.

Ticket 5. Oscillation and the oscillation criterion

Core idea. The obstruction to Riemann integrability is persistent oscillation on sets of non-negligible total length.

Definition 5.1 (Oscillation)**.** *For a bounded function $f : E \to \mathbb{R}$, define its oscillation on E by*

$$\operatorname{osc}_E f = \sup_E f - \inf_E f.$$

For a partition $P : a = x_0 < \cdots < x_n = b$, define

$$\Omega(f,P) = \sum_{i=1}^{n} \operatorname{osc}_{[x_{i-1},x_i]} f \,(x_i - x_{i-1}).$$

Proposition 5.1 (Oscillation equals Darboux gap)**.** *For every bounded f and partition P,*

$$U(f,P) - L(f,P) = \Omega(f,P).$$

Proof. By definition,

$$U(f,P) - L(f,P) = \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}),$$

where $M_i = \sup_{[x_{i-1},x_i]} f$ and $m_i = \inf_{[x_{i-1},x_i]} f$. But $M_i - m_i = \operatorname{osc}_{[x_{i-1},x_i]} f$. This is exactly $\Omega(f,P)$. □

Theorem 5.1 (Oscillation criterion)**.** *A bounded function $f : [a,b] \to \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that*

$$\Omega(f,P) < \varepsilon.$$

Proof. By the previous proposition, $\Omega(f,P) = U(f,P) - L(f,P)$. Therefore the result is exactly the Darboux criterion for integrability. □

Definition 5.2 (Pointwise oscillation)**.** *The oscillation of f at x is*

$$\omega_f(x) = \inf_{\delta>0} \operatorname{osc}_{[a,b]\cap(x-\delta,x+\delta)} f.$$

The function f is continuous at x if and only if $\omega_f(x) = 0$.

Proposition 5.2 (Continuity and pointwise oscillation)**.** *For bounded f, f is continuous at x if and only if $\omega_f(x) = 0$.*

Proof. If f is continuous at x, then for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(y) - f(x)| < \varepsilon/2$ whenever $|y - x| < \delta$. Thus any two values in the neighborhood differ by less than ε, so the local oscillation is less than ε. Hence $\omega_f(x) = 0$.

Conversely, if $\omega_f(x) = 0$, then for every $\varepsilon > 0$ some neighborhood of x has oscillation less than ε. Since x is in that neighborhood, for all nearby y, $|f(y) - f(x)| < \varepsilon$. Hence f is continuous at x. □

Generalized viewpoint. Oscillation is the seminorm-like local quantity that measures failure of continuity. The integral sees a weighted average of local oscillations over a partition.

Practice problems.

1. Compute $\Omega(f,P)$ for a monotone function and recover the proof of monotone integrability.

2. Prove that a function with finitely many discontinuities is integrable using oscillations.

3. Show that the Dirichlet function $1_{\mathbb{Q}}$ on $[0,1]$ is not Riemann integrable.

Ticket 6. Cantor theorem and integrability of continuous functions

Core idea. Continuity is local. Compactness turns local control into uniform control. Uniform control makes Darboux oscillation small.

Theorem 6.1 (Cantor–Heine theorem)**.** *Let K be a compact metric space and $f : K \to \mathbb{R}$ be continuous. Then f is uniformly continuous: for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$d(x,y) < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon$$

for all $x, y \in K$.

Proof. For each $x \in K$, continuity gives $r_x > 0$ such that

$$d(x,y) < r_x \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon/2.$$

The balls $B(x, r_x/2)$ cover K. By compactness, choose a finite subcover

$$K \subset B(x_1, r_{x_1}/2) \cup \cdots \cup B(x_N, r_{x_N}/2).$$

Let $\delta = \min_i r_{x_i}/2$. If $d(y,z) < \delta$, choose i such that $y \in B(x_i, r_{x_i}/2)$. Then

$$d(z, x_i) \le d(z,y) + d(y, x_i) < \delta + r_{x_i}/2 \le r_{x_i}.$$

Also $d(y, x_i) < r_{x_i}$. Hence

$$|f(y) - f(z)| \le |f(y) - f(x_i)| + |f(x_i) - f(z)| < \varepsilon.$$

Thus f is uniformly continuous. □

Theorem 6.2 (Continuous functions on compact intervals are integrable)**.** *If $f : [a,b] \to \mathbb{R}$ is continuous, then f is Riemann integrable.*

Proof. By Cantor's theorem, f is uniformly continuous on $[a,b]$. Given $\varepsilon > 0$, choose $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon/(b-a)$. If P has mesh less than δ, then on every subinterval $[x_{i-1}, x_i]$ the oscillation of f is at most $\varepsilon/(b-a)$. Therefore

$$U(f,P) - L(f,P) = \sum_i \operatorname{osc}_{[x_{i-1},x_i]} f \,(x_i - x_{i-1}) \le \frac{\varepsilon}{b-a} \sum_i (x_i - x_{i-1}) = \varepsilon.$$

By the Darboux criterion, f is integrable. □

Practice problems.

1. Prove Cantor's theorem by contradiction using sequences, and compare it with the finite-subcover proof.
2. Use uniform continuity to prove integrability of continuous functions on $[a,b]$ without mentioning Darboux sums explicitly until the final step.
3. Give a continuous function on an open interval that is not uniformly continuous. Explain why compactness is missing.

Ticket 7. Algebra of integrable functions

Core idea. Integrable functions are stable under algebraic operations because oscillation estimates are stable under addition, scalar multiplication, multiplication, and reciprocal operations away from zero.

Theorem 7.1 (Linear combinations)**.** *If f, g are Riemann integrable on $[a,b]$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha f + \beta g$ is integrable and*

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g.$$

Proof. For any subinterval I,

$$\operatorname{osc}_I(\alpha f + \beta g) \le |\alpha| \operatorname{osc}_I f + |\beta| \operatorname{osc}_I g.$$

Indeed, for $x, y \in I$, take the supremum of

$$|\alpha(f(x) - f(y)) + \beta(g(x) - g(y))|.$$

Therefore

$$\Omega(\alpha f + \beta g, P) \le |\alpha|\Omega(f, P) + |\beta|\Omega(g, P).$$

Choose partitions making the right side arbitrarily small, using a common refinement. Hence $\alpha f + \beta g$ is integrable.

To prove the formula for the integral, use Riemann sums:

$$S(\alpha f + \beta g, P, \xi) = \alpha S(f, P, \xi) + \beta S(g, P, \xi).$$

Passing to the fine-partition limit gives the formula. □

Theorem 7.2 (Products)**.** *If f, g are integrable on $[a, b]$, then fg is integrable.*

Proof. Since integrable functions are bounded, choose $|f| \le A$, $|g| \le B$. For $x, y \in I$,

$$|f(x)g(x) - f(y)g(y)| \le |f(x)|\,|g(x) - g(y)| + |g(y)|\,|f(x) - f(y)| \le A \operatorname{osc}_I g + B \operatorname{osc}_I f.$$

Hence

$$\Omega(fg, P) \le A\Omega(g, P) + B\Omega(f, P).$$

Using the integrability of f and g, choose a common refinement for which the right side is arbitrarily small. Thus fg is integrable. □

Theorem 7.3 (Reciprocal and quotient)**.** *If g is integrable and $|g(x)| \ge c > 0$ on $[a, b]$, then $1/g$ is integrable. If additionally f is integrable, then f/g is integrable.*

Proof. For $x, y \in I$,

$$\left|\frac{1}{g(x)} - \frac{1}{g(y)}\right| = \frac{|g(x) - g(y)|}{|g(x)g(y)|} \le \frac{1}{c^2}|g(x) - g(y)|.$$

Thus $\operatorname{osc}_I(1/g) \le c^{-2} \operatorname{osc}_I g$, and $1/g$ is integrable by the oscillation criterion. The quotient $f/g = f \cdot (1/g)$ is then integrable by the product theorem. □

Practice problems.

1. Prove the oscillation inequality for $\alpha f + \beta g$ in full detail.

2. Let f be integrable. Prove that f^2 is integrable as a special case of the product theorem.

3. Give an example showing why the condition $|g| \ge c > 0$ is necessary for the quotient theorem.

Ticket 8. Monotonicity of the integral, absolute value, and finite changes

Core idea. The integral is an order-preserving linear functional on the space of integrable functions, and it ignores finite sets.

Theorem 8.1 (Order monotonicity)**.** *If f, g are integrable on $[a, b]$ and $f(x) \le g(x)$ for all x, then*

$$\int_a^b f \le \int_a^b g.$$

In particular, if $m \le f(x) \le M$, then

$$m(b-a) \le \int_a^b f \le M(b-a).$$

Proof. For any tagged partition,

$$S(f, P, \xi) \le S(g, P, \xi).$$

Taking the fine-partition limit gives the first inequality. The second follows by comparing f with the constant functions m and M. ☐

Theorem 8.2 (Absolute value)**.** *If f is integrable, then $|f|$ is integrable and*

$$\left| \int_a^b f \right| \le \int_a^b |f|.$$

Proof. For any interval I,

$$\operatorname{osc}_I |f| \le \operatorname{osc}_I f,$$

because $||u| - |v|| \le |u - v|$. Hence $\Omega(|f|, P) \le \Omega(f, P)$, so $|f|$ is integrable. Since $-|f| \le f \le |f|$, monotonicity gives

$$-\int_a^b |f| \le \int_a^b f \le \int_a^b |f|,$$

which is equivalent to the desired inequality. ☐

Theorem 8.3 (Changing finitely many values)**.** *If f is integrable and g differs from f at only finitely many points, then g is integrable and*

$$\int_a^b g = \int_a^b f.$$

Proof. It suffices to prove the result when $h = g - f$ is zero except at finitely many points. Since finite-valued modifications of a bounded integrable function are bounded if the modified values are finite, let $|h| \le M$. Given $\varepsilon > 0$, cover the finitely many nonzero points of h by finitely many intervals whose total length is less than $\varepsilon/(2M)$. Construct a partition containing the endpoints of these intervals. On subintervals not meeting these points, the oscillation of h is zero; on the exceptional subintervals, it is at most $2M$. Thus

$$\Omega(h, P) \le 2M \cdot \frac{\varepsilon}{2M} = \varepsilon.$$

Therefore h is integrable. Since every Riemann sum of h can be made arbitrarily small by isolating the exceptional points, its integral is 0. Hence $g = f + h$ is integrable and has the same integral as f. ☐

Practice problems.

1. Prove that if $f \ge 0$ is integrable, then $\int_a^b f \ge 0$.

2. Prove $|\int f| \le \int |f|$ using only order monotonicity.

3. Let $g = 0$ except at finitely many points. Prove directly from Darboux sums that $\int g = 0$.

Ticket 9. Additivity over intervals and the mean value theorem

Core idea. The integral is local with respect to decomposition of the interval. Additivity is the finite-measure analogue of finite additivity of a measure.

Theorem 9.1 (Additivity over intervals). *Let $a < c < b$. A bounded function $f : [a, b] \to \mathbb{R}$ is integrable on $[a, b]$ if and only if it is integrable on $[a, c]$ and $[c, b]$. In that case,*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof. If f is integrable on $[a, b]$, then given $\varepsilon > 0$, choose a partition P of $[a, b]$ with $U(f, P) - L(f, P) < \varepsilon$ and refine it by inserting c. The restrictions of the refined partition to $[a, c]$ and $[c, b]$ have Darboux gaps whose sum is less than ε, so both restrictions are integrable.

Conversely, if f is integrable on the two subintervals, choose partitions P_1 and P_2 with Darboux gaps less than $\varepsilon/2$ on each part. Their union is a partition of $[a, b]$ with total gap less than ε, so f is integrable on $[a, b]$.

For the formula, take tagged partitions on the two subintervals and concatenate them. Riemann sums add exactly, and passing to the limit gives the integral identity. □

Theorem 9.2 (Integrability on smaller intervals). *If f is integrable on $[a, b]$, then f is integrable on every subinterval $[\alpha, \beta] \subset [a, b]$.*

Proof. Apply the previous theorem twice: first split at α, then at β. □

Theorem 9.3 (Mean value theorem for integrals). *If $f : [a, b] \to \mathbb{R}$ is continuous, then there exists $c \in [a, b]$ such that*

$$\int_a^b f(x)\,dx = f(c)(b - a).$$

Proof. By the extreme value theorem, f attains a minimum m and maximum M on $[a, b]$. Monotonicity gives

$$m(b - a) \leq \int_a^b f \leq M(b - a).$$

Hence

$$A = \frac{1}{b - a} \int_a^b f$$

lies in $[m, M]$. Since $f([a, b])$ is an interval by the intermediate value theorem, there exists $c \in [a, b]$ with $f(c) = A$. This is the desired formula. □

Practice problems.

1. Prove additivity for three subintervals and then for a finite partition of $[a, b]$.
2. Show that if f is integrable on $[a, b]$, then $x \mapsto \int_a^x f$ is well-defined for all $x \in [a, b]$.
3. Prove the mean value theorem for integrals when f is continuous and nonnegative.

Ticket 10. Variable upper limit and the Newton–Leibniz formula

Core idea. The map $x \mapsto \int_a^x f$ is the bridge between integration and differentiation.

Definition 10.1 (Integral with variable upper limit)**.** *Let f be integrable on $[a,b]$. Define*

$$F(x) = \int_a^x f(t)\,dt, \qquad x \in [a,b].$$

Theorem 10.1 (Continuity of the variable integral)**.** *If f is integrable on $[a,b]$, then $F(x) = \int_a^x f$ is continuous on $[a,b]$.*

Proof. Since f is integrable, it is bounded; let $|f| \le M$. For $x, y \in [a,b]$, assume $x < y$. By additivity and monotonicity,

$$|F(y) - F(x)| = \left| \int_x^y f(t)\,dt \right| \le \int_x^y |f(t)|\,dt \le M(y-x).$$

Thus $|F(y) - F(x)| \le M|y-x|$, so F is Lipschitz and hence continuous. □

Theorem 10.2 (Differentiability at points of continuity)**.** *If f is integrable on $[a,b]$ and continuous at $x \in (a,b)$, then $F(x) = \int_a^x f$ is differentiable at x, and*

$$F'(x) = f(x).$$

Proof. For $h \ne 0$ small enough,

$$\frac{F(x+h) - F(x)}{h} - f(x) = \frac{1}{h} \int_x^{x+h} (f(t) - f(x))\,dt.$$

Given $\varepsilon > 0$, by continuity of f at x, choose $\delta > 0$ such that $|t - x| < \delta$ implies $|f(t) - f(x)| < \varepsilon$. If $|h| < \delta$, then all t between x and $x+h$ satisfy this inequality. Therefore

$$\left| \frac{1}{h} \int_x^{x+h} (f(t) - f(x))\,dt \right| \le \frac{1}{|h|} \int_{\min(x,x+h)}^{\max(x,x+h)} \varepsilon\,dt = \varepsilon.$$

Thus the difference quotient tends to $f(x)$. □

Theorem 10.3 (Newton–Leibniz formula)**.** *If f is continuous on $[a,b]$ and Φ is any primitive of f, then*

$$\int_a^b f(x)\,dx = \Phi(b) - \Phi(a).$$

Proof. By the previous theorem, $F(x) = \int_a^x f$ is a primitive of f. Since Φ is also a primitive, $\Phi - F$ is constant on $[a,b]$. Therefore

$$\Phi(b) - F(b) = \Phi(a) - F(a).$$

But $F(a) = 0$ and $F(b) = \int_a^b f$. Hence

$$\int_a^b f = F(b) = \Phi(b) - \Phi(a).$$

□

Practice problems.

1. Prove that $F(x) = \int_a^x f$ is Lipschitz when f is bounded.
2. Give an example where f is integrable but discontinuous at x_0, and F' fails to equal $f(x_0)$.
3. Use Newton–Leibniz to compute $\int_0^1 x^n\,dx$ and justify the primitive.

Ticket 11. Definite integration by parts, substitution, and Taylor's formula

Core idea. The classical calculus rules are consequences of product rule, chain rule, and Newton–Leibniz.

Theorem 11.1 (Integration by parts)**.** *If $u, v \in C^1([a,b])$, then*

$$\int_a^b u'(x)v(x)\,dx = u(b)v(b) - u(a)v(a) - \int_a^b u(x)v'(x)\,dx.$$

Proof. By the product rule, $(uv)' = u'v + uv'$. Integrating both sides and applying Newton–Leibniz,

$$\int_a^b u'v + \int_a^b uv' = \int_a^b (uv)' = u(b)v(b) - u(a)v(a).$$

Rearranging gives the formula. □

Theorem 11.2 (Change of variables)**.** *Let $\varphi : [\alpha, \beta] \to [a, b]$ be continuously differentiable, and let f be continuous on an interval containing $\varphi([\alpha, \beta])$. Then*

$$\int_\alpha^\beta f(\varphi(t))\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx.$$

Proof. Let F be a primitive of f, which exists by the fundamental theorem. By the chain rule,

$$(F \circ \varphi)'(t) = f(\varphi(t))\varphi'(t).$$

Applying Newton–Leibniz on $[\alpha, \beta]$,

$$\int_\alpha^\beta f(\varphi(t))\varphi'(t)\,dt = F(\varphi(\beta)) - F(\varphi(\alpha)) = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx.$$

□

Theorem 11.3 (Taylor formula with integral remainder)**.** *Let $f \in C^{n+1}([a,b])$, and fix a. Then for every $x \in [a,b]$,*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt.$$

Proof. We prove by induction on n. For $n = 0$, the formula is Newton–Leibniz:

$$f(x) = f(a) + \int_a^x f'(t)\,dt.$$

Assume the formula holds for $n-1$ applied to f:

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{1}{(n-1)!}\int_a^x f^{(n)}(t)(x-t)^{n-1}\,dt.$$

Integrate the remainder by parts with

$$u = f^{(n)}(t), \qquad dv = (x-t)^{n-1}\,dt.$$

Since $v = -(x-t)^n/n$,

$$\int_a^x f^{(n)}(t)(x-t)^{n-1}\,dt = \frac{f^{(n)}(a)}{n}(x-a)^n + \frac{1}{n}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt.$$

Dividing by $(n-1)!$ gives the desired formula. □

Practice problems.

1. Derive the formula $\int_a^b uv' = u(b)v(b) - u(a)v(a) - \int_a^b u'v$.
2. Prove the change-of-variables formula first for increasing φ, then explain why the primitive proof also covers decreasing φ.
3. Write Taylor's formula with integral remainder for $n = 2$ and verify it by differentiating the right-hand side.

Ticket 12. Paths in $\mathbb{R}^d$: concatenation, equivalence, and length

Core idea. Path length is defined as a supremum over polygonal approximations. This makes it invariant under orientation-preserving reparametrization.

Definition 12.1 (Path and concatenation)**.** *A path in $\mathbb{R}^d$ is a continuous map $\gamma : [a, b] \to \mathbb{R}^d$. If $\gamma : [a, b] \to \mathbb{R}^d$ and $\eta : [b, c] \to \mathbb{R}^d$ satisfy $\gamma(b) = \eta(b)$, their concatenation is the path $\gamma * \eta : [a, c] \to \mathbb{R}^d$ equal to γ on $[a, b]$ and η on $[b, c]$.*

Definition 12.2 (Equivalent paths)**.** *Two paths $\gamma : [a, b] \to \mathbb{R}^d$ and $\eta : [c, d] \to \mathbb{R}^d$ are equivalent if there exists an increasing homeomorphism $\varphi : [c, d] \to [a, b]$ such that*

$$\eta = \gamma \circ \varphi.$$

This is an orientation-preserving reparametrization.

Definition 12.3 (Length)**.** *The length of a path $\gamma : [a, b] \to \mathbb{R}^d$ is*

$$\ell(\gamma) = \sup_P \sum_{i=1}^{n} \|\gamma(t_i) - \gamma(t_{i-1})\|,$$

where the supremum is over all partitions $P : a = t_0 < \cdots < t_n = b$. If the supremum is finite, γ is rectifiable.

Theorem 12.1 (Invariance under reparametrization)**.** *Equivalent paths have equal length.*

Proof. Let $\eta = \gamma \circ \varphi$, where $\varphi : [c, d] \to [a, b]$ is an increasing homeomorphism. Given a partition $Q : c = s_0 < \cdots < s_n = d$, the points $t_i = \varphi(s_i)$ form a partition of $[a, b]$. Therefore

$$\sum_i \|\eta(s_i) - \eta(s_{i-1})\| = \sum_i \|\gamma(t_i) - \gamma(t_{i-1})\| \le \ell(\gamma).$$

Taking the supremum over Q gives $\ell(\eta) \le \ell(\gamma)$. Applying the same argument to $\gamma = \eta \circ \varphi^{-1}$ gives the reverse inequality. □

Theorem 12.2 (Length of concatenation)**.** *If γ and η are composable paths, then*

$$\ell(\gamma * \eta) = \ell(\gamma) + \ell(\eta).$$

Proof. Any partition of the concatenated interval can be refined by inserting the joining point. Refinement cannot decrease polygonal length because of the triangle inequality. Thus the supremum may be taken over partitions containing the joining point. For such partitions, the polygonal sum splits exactly into a sum for γ and a sum for η. Taking suprema gives the formula. □

Practice problems.

1. Prove that refinement of a partition cannot decrease the polygonal length of a path.
2. Show that the equivalence relation on paths is reflexive, symmetric, and transitive.
3. Compute the length of the polygonal path through points $p_0, p_1, \dots, p_n \in \mathbb{R}^d$.

Ticket 13. Length of a smooth path

Core idea. For smooth paths, abstract length as a supremum equals the integral of speed.

Definition 13.1 (Smooth path)**.** *A path $\gamma : [a, b] \to \mathbb{R}^d$ is C^1 if all coordinate functions are continuously differentiable. Its velocity is $\gamma'(t)$, and its speed is $\|\gamma'(t)\|$.*

Theorem 13.1 (Length formula for C^1 paths)**.** *If $\gamma \in C^1([a, b], \mathbb{R}^d)$, then γ is rectifiable and*

$$\ell(\gamma) = \int_a^b \|\gamma'(t)\| \, dt.$$

Proof. For any partition $P : a = t_0 < \cdots < t_n = b$, Newton–Leibniz applied coordinatewise gives

$$\gamma(t_i) - \gamma(t_{i-1}) = \int_{t_{i-1}}^{t_i} \gamma'(t) \, dt.$$

Hence

$$\|\gamma(t_i) - \gamma(t_{i-1})\| \le \int_{t_{i-1}}^{t_i} \|\gamma'(t)\| \, dt.$$

Summing over i gives every polygonal sum $\le \int_a^b \|\gamma'\|$, so $\ell(\gamma) \le \int_a^b \|\gamma'\|$.

For the reverse inequality, use uniform continuity of γ' on $[a, b]$. Given $\varepsilon > 0$, choose $\delta > 0$ such that $|s - t| < \delta$ implies $\|\gamma'(s) - \gamma'(t)\| < \varepsilon/(b - a)$. Let P have mesh less than δ. For each interval choose $\tau_i \in [t_{i-1}, t_i]$. Then

$$\int_{t_{i-1}}^{t_i} \|\gamma'(t)\| \, dt \le \|\gamma'(\tau_i)\|(t_i - t_{i-1}) + \frac{\varepsilon}{b - a}(t_i - t_{i-1}).$$

Also

$$\gamma(t_i) - \gamma(t_{i-1}) = \gamma'(\tau_i)(t_i - t_{i-1}) + r_i,$$

where

$$\|r_i\| \le \int_{t_{i-1}}^{t_i} \|\gamma'(t) - \gamma'(\tau_i)\| \, dt \le \frac{\varepsilon}{b - a}(t_i - t_{i-1}).$$

Thus

$$\|\gamma'(\tau_i)\|(t_i - t_{i-1}) \le \|\gamma(t_i) - \gamma(t_{i-1})\| + \frac{\varepsilon}{b - a}(t_i - t_{i-1}).$$

Combining and summing yields

$$\int_a^b \|\gamma'\| \le \sum_i \|\gamma(t_i) - \gamma(t_{i-1})\| + 2\varepsilon \le \ell(\gamma) + 2\varepsilon.$$

Since ε is arbitrary, $\int_a^b \|\gamma'\| \le \ell(\gamma)$. □

Practice problems.

1. Compute the length of $\gamma(t) = (R \cos t, R \sin t)$, $t \in [0, 2\pi]$.
2. Prove that if $\gamma'(t) = 0$ for all t, then $\ell(\gamma) = 0$ and γ is constant.
3. Show that the length formula is invariant under a C^1 increasing reparametrization.

Ticket 14. Improper integrals and the Cauchy criterion

Core idea. An improper integral is a limit of proper integrals over an exhaustion of the domain. Its convergence is therefore a Cauchy property.

Definition 14.1 (Improper integral on a half-interval). *Let f be Riemann integrable on every $[a, A]$, $A > a$. Define*

$$\int_a^\infty f(x)\,dx = \lim_{A\to\infty} \int_a^A f(x)\,dx,$$

if this finite limit exists. Similar definitions apply to singular endpoints, for example

$$\int_a^b f = \lim_{\alpha\downarrow a} \int_\alpha^b f$$

when f is integrable on every $[\alpha, b]$, $\alpha > a$.

Theorem 14.1 (Agreement with the Riemann integral). *If f is Riemann integrable on $[a, b]$, then the improper integral over $[a, b]$, interpreted by endpoint limits, equals the ordinary Riemann integral.*

Proof. Let $F(x) = \int_a^x f$. Since f is integrable, F is continuous. Hence

$$\lim_{\beta\uparrow b} \int_a^\beta f = \lim_{\beta\uparrow b} F(\beta) = F(b) = \int_a^b f.$$

The argument at the left endpoint is analogous. □

Theorem 14.2 (Cauchy criterion). *The improper integral $\int_a^\infty f$ converges if and only if for every $\varepsilon > 0$ there exists A_0 such that for all $B > C \geq A_0$,*

$$\left|\int_C^B f(x)\,dx\right| < \varepsilon.$$

Proof. Let $F(A) = \int_a^A f$. The improper integral converges exactly when $F(A)$ has a finite limit as $A \to \infty$. Since $\mathbb{R}$ is complete, this is equivalent to the Cauchy condition: for every $\varepsilon > 0$, $|F(B) - F(C)| < \varepsilon$ for all sufficiently large B, C. But

$$F(B) - F(C) = \int_C^B f.$$

This proves the criterion. □

Theorem 14.3 (Additivity and convergence on smaller intervals). *If $\int_a^\infty f$ converges, then for every $c > a$, $\int_c^\infty f$ converges and*

$$\int_a^\infty f = \int_a^c f + \int_c^\infty f.$$

Proof. For $A > c$, ordinary additivity gives

$$\int_a^A f = \int_a^c f + \int_c^A f.$$

Letting $A \to \infty$, the left side converges by assumption, so $\int_c^A f$ has a finite limit and the formula follows. □

Theorem 14.4 (Improper integration by parts)**.** *Suppose $u, v \in C^1([a,\infty))$, and assume the finite limit $\lim_{A\to\infty} u(A)v(A)$ exists and one of the improper integrals $\int_a^\infty u'v$, $\int_a^\infty uv'$ converges. Then the other converges if and only if the integration-by-parts identity has a finite right-hand side, and*

$$\int_a^\infty u'v = [uv]_a^\infty - \int_a^\infty uv'.$$

Proof. For every finite $A > a$, proper integration by parts gives

$$\int_a^A u'v = u(A)v(A) - u(a)v(a) - \int_a^A uv'.$$

Taking $A \to \infty$ proves the statement whenever the relevant limits exist. Conversely, if two of the three terms have finite limits, the third must also have a finite limit by rearranging the equality. □

Practice problems.

1. Prove the Cauchy criterion for an improper integral with a singularity at a.
2. Determine for which p the integral $\int_1^\infty x^{-p}\,dx$ converges.
3. Prove additivity for $\int_a^b f$ when both endpoints are improper.

Ticket 15. Absolute convergence and comparison tests for improper integrals

Core idea. Absolute convergence converts cancellation-dependent convergence into ordinary Cauchy smallness of tails.

Definition 15.1 (Absolute convergence)**.** *An improper integral $\int_a^\infty f$ is absolutely convergent if $\int_a^\infty |f|$ converges.*

Theorem 15.1 (Absolute convergence implies convergence)**.** *If $\int_a^\infty |f|$ converges, then $\int_a^\infty f$ converges.*

Proof. By the Cauchy criterion for $\int |f|$, for every $\varepsilon > 0$ there exists A_0 such that for $B > C \geq A_0$,

$$\int_C^B |f| < \varepsilon.$$

Then

$$\left| \int_C^B f \right| \leq \int_C^B |f| < \varepsilon.$$

The Cauchy criterion for $\int f$ gives convergence. □

Theorem 15.2 (Comparison test)**.** *Suppose $0 \leq |f(x)| \leq g(x)$ for all sufficiently large x, and $\int_a^\infty g$ converges. Then $\int_a^\infty f$ converges absolutely.*

Proof. Choose A such that $|f| \leq g$ on $[A,\infty)$. For $B > A$, monotonicity of the proper integral gives

$$0 \leq \int_A^B |f| \leq \int_A^B g \leq \int_A^\infty g.$$

The increasing function $B \mapsto \int_A^B |f|$ is bounded above, so it has a finite limit. Therefore $\int_A^\infty |f|$ converges, and adding the finite integral over $[a, A]$ gives absolute convergence. □

Corollary 15.1 (Limit comparison)**.** *Let $f, g \geq 0$ eventually and suppose*

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = L, \qquad 0 < L < \infty.$$

Then $\int_a^\infty f$ and $\int_a^\infty g$ either both converge or both diverge.

Proof. For sufficiently large x, say $A \leq x$, the limit condition gives

$$\frac{L}{2} g(x) \leq f(x) \leq \frac{3L}{2} g(x).$$

The comparison test in both directions proves the claim. □

Practice problems.

1. Prove that $\int_1^\infty \frac{\sin x}{x^p}\, dx$ is absolutely convergent for $p > 1$.
2. Use limit comparison to determine convergence of $\int_2^\infty \frac{dx}{x \log^p x}$.
3. Give an example of a convergent improper integral that is not absolutely convergent.

Ticket 16. Dirichlet and Abel tests for improper integrals

Core idea. Dirichlet and Abel tests control cancellation by integrating a bounded primitive against a monotone factor.

Theorem 16.1 (Dirichlet test for improper integrals)**.** *Let f be continuous on $[a, \infty)$, and suppose its primitive integrals*

$$F(B) = \int_a^B f(x)\, dx$$

are bounded. Let $g \in C^1([a, \infty))$ be monotone, $g(x) \to 0$, and g bounded. Then

$$\int_a^\infty f(x) g(x)\, dx$$

converges.

Proof. For $B > C$, integration by parts on $[C, B]$ gives

$$\int_C^B fg = F(B)g(B) - F(C)g(C) - \int_C^B F(x) g'(x)\, dx,$$

where $F(x) = \int_a^x f$. Let $|F| \leq M$. Since g is monotone and tends to 0, its total variation on $[C, B]$ is at most $|g(C)| + |g(B)| \leq 2|g(C)|$ for large C, more simply if $g \geq 0$ decreasing then $\int_C^B |g'| = g(C) - g(B) \leq g(C)$. In general apply the same argument to $|g|$ after the eventual sign is fixed by convergence to 0 and monotonicity. Hence

$$\left| \int_C^B fg \right| \leq M|g(B)| + M|g(C)| + M \int_C^B |g'| \to 0$$

as $C, B \to \infty$. By the Cauchy criterion, the integral converges. □

Theorem 16.2 (Abel test for improper integrals)**.** *If $\int_a^\infty f$ converges and $g \in C^1([a, \infty))$ is monotone and bounded, then $\int_a^\infty fg$ converges.*

Proof. Since g is monotone and bounded, $g(x) \to L$ for some finite L. Write

$$fg = f(g - L) + Lf.$$

The integral of Lf converges by assumption. For $f(g - L)$, the primitive $F(B) = \int_a^B f$ is convergent, hence bounded, while $g - L$ is monotone and tends to 0. Dirichlet's test applies. ☐

Practice problems.

1. Use Dirichlet's test to prove convergence of $\int_1^\infty \frac{\sin x}{x}\,dx$.
2. Prove Abel's test from Dirichlet's test in full detail.
3. Find an example where Dirichlet gives convergence but absolute convergence fails.

Ticket 17. Numerical series: Cauchy criterion, geometric series, and basic properties

Core idea. A numerical series is a sequence of partial sums. All convergence questions are therefore questions about convergence in $\mathbb{R}$ or $\mathbb{C}$.

Definition 17.1 (Series and partial sums)**.** *For a sequence (a_n) in $\mathbb{K}$, the series $\sum_{n=1}^\infty a_n$ is the sequence of partial sums*

$$s_N = \sum_{n=1}^N a_n.$$

The series converges to s if $s_N \to s$.

Theorem 17.1 (Cauchy criterion for series)**.** *The series $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there exists N such that for all $m > n \geq N$,*

$$\left| \sum_{k=n+1}^m a_k \right| < \varepsilon.$$

Proof. The partial sums (s_N) converge in the complete space $\mathbb{K}$ if and only if they are Cauchy. But

$$s_m - s_n = \sum_{k=n+1}^m a_k.$$

This is exactly the stated condition. ☐

Theorem 17.2 (Geometric series)**.** *For $q \in \mathbb{K}$, the series $\sum_{n=0}^\infty q^n$ converges if and only if $|q| < 1$. In that case,*

$$\sum_{n=0}^\infty q^n = \frac{1}{1-q}.$$

Proof. For $q \neq 1$, finite summation gives

$$\sum_{n=0}^N q^n = \frac{1 - q^{N+1}}{1-q}.$$

If $|q| < 1$, then $q^{N+1} \to 0$, so the sum tends to $1/(1-q)$. If $|q| \geq 1$, the terms q^n do not tend to 0 when $|q| > 1$, and for $|q| = 1$ they also fail to tend to 0 except impossible cancellation of individual terms; in all cases the necessary condition $a_n \to 0$ fails. Thus the series diverges. ☐

Proposition 17.1 (Linearity and tails). *If $\sum a_n$ and $\sum b_n$ converge, then $\sum(\alpha a_n + \beta b_n)$ converges and*

$$\sum(\alpha a_n + \beta b_n) = \alpha \sum a_n + \beta \sum b_n.$$

Also, convergence is unaffected by changing or deleting finitely many initial terms.

Proof. Both claims follow from the corresponding facts for limits of partial sums. Finite changes alter partial sums by a fixed constant after some index, so they affect only the value by a finite amount, not convergence of the tail. □

Practice problems.

1. Prove that convergence of $\sum a_n$ implies $a_n \to 0$.
2. Compute $\sum_{n=2}^{\infty} 3/2^n$ using the geometric series formula.
3. Use the Cauchy criterion to prove divergence of the harmonic series.

Ticket 18. Absolute convergence and root/ratio tests for series

Core idea. Absolute convergence is convergence in the norm before summing signs or phases. It is the discrete analogue of absolute convergence for improper integrals.

Definition 18.1 (Absolute convergence). *A series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges.*

Theorem 18.1 (Absolute convergence implies convergence). *If $\sum |a_n|$ converges, then $\sum a_n$ converges.*

Proof. For $m > n$,

$$\left| \sum_{k=n+1}^{m} a_k \right| \leq \sum_{k=n+1}^{m} |a_k|.$$

Since $\sum |a_n|$ converges, the right-hand tails tend to 0. The Cauchy criterion gives convergence of $\sum a_n$. □

Theorem 18.2 (Comparison test). *If $|a_n| \leq b_n$ eventually and $\sum b_n$ converges with $b_n \geq 0$, then $\sum a_n$ converges absolutely.*

Proof. The tails of $\sum |a_n|$ are bounded by the corresponding tails of $\sum b_n$, which tend to 0. Apply the Cauchy criterion. □

Theorem 18.3 (Root test). *Let*

$$L = \limsup_{n\to\infty} |a_n|^{1/n}.$$

If $L < 1$, then $\sum a_n$ converges absolutely. If $L > 1$, then the series diverges.

Proof. If $L < 1$, choose q with $L < q < 1$. Then for all sufficiently large n, $|a_n|^{1/n} \leq q$, so $|a_n| \leq q^n$. The geometric comparison test gives absolute convergence.

If $L > 1$, infinitely many n satisfy $|a_n|^{1/n} > 1$, hence $|a_n| > 1$ infinitely often. Thus $a_n \not\to 0$, so the series diverges. □

Theorem 18.4 (Ratio test). *If $a_n \neq 0$ eventually and*

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L,$$

then $\sum a_n$ converges absolutely for $L < 1$ and diverges for $L > 1$.

Proof. If $L < 1$, choose q with $L < q < 1$. For large n, $|a_{n+1}| \leq q|a_n|$. Iterating gives $|a_{N+k}| \leq |a_N|q^k$, so the tail is dominated by a geometric series.

If $L > 1$, then for large n, $|a_{n+1}| \geq q|a_n|$ for some $q > 1$, so $|a_n|$ does not tend to zero. Therefore the series diverges. $\square$

Practice problems.

1. Apply the root test to $\sum n^3/2^n$.

2. Apply the ratio test to $\sum n!/n^n$.

3. Give an example where both root and ratio tests are inconclusive but the series converges.

Ticket 19. Integral test, Dirichlet test, and Abel test for series

Core idea. Series tests are discrete versions of improper-integral tests. Dirichlet and Abel are governed by summation by parts.

Theorem 19.1 (Integral test)**.** *Let $f : [1, \infty) \to [0, \infty)$ be decreasing. Then $\sum_{n=1}^{\infty} f(n)$ converges if and only if $\int_1^{\infty} f(x)\,dx$ converges.*

Proof. For decreasing f, for $x \in [n, n+1]$,

$$f(n+1) \leq f(x) \leq f(n).$$

Integrating over $[n, n+1]$ gives

$$f(n+1) \leq \int_n^{n+1} f(x)\,dx \leq f(n).$$

Summing from $n = 1$ to N yields

$$\sum_{n=2}^{N+1} f(n) \leq \int_1^{N+1} f(x)\,dx \leq \sum_{n=1}^{N} f(n).$$

Since all terms are nonnegative, boundedness of one increasing sequence of partial sums is equivalent to boundedness of the other. Thus convergence is equivalent. $\square$

Lemma 19.1 (Abel summation)**.** *Let $A_n = \sum_{k=1}^{n} a_k$. Then for $m > n$,*

$$\sum_{k=n+1}^{m} a_k b_k = A_m b_m - A_n b_{n+1} + \sum_{k=n+1}^{m-1} A_k (b_k - b_{k+1}).$$

Proof. Since $a_k = A_k - A_{k-1}$,

$$\sum_{k=n+1}^{m} a_k b_k = \sum_{k=n+1}^{m} (A_k - A_{k-1}) b_k.$$

Expanding and shifting indices gives a telescoping identity:

$$\sum_{k=n+1}^{m} A_k b_k - \sum_{k=n+1}^{m} A_{k-1} b_k = A_m b_m - A_n b_{n+1} + \sum_{k=n+1}^{m-1} A_k (b_k - b_{k+1}).$$

$\square$

Theorem 19.2 (Dirichlet test for series). *If the partial sums $A_n = \sum_{k=1}^{n} a_k$ are bounded, and b_n is monotone with $b_n \to 0$, then $\sum a_n b_n$ converges.*

Proof. Assume $|A_n| \le M$. By Abel summation, for $m > n$,

$$\left| \sum_{k=n+1}^{m} a_k b_k \right| \le M|b_m| + M|b_{n+1}| + M \sum_{k=n+1}^{m-1} |b_k - b_{k+1}|.$$

Since (b_n) is monotone and tends to 0, the total variation tail $\sum |b_k - b_{k+1}|$ from $n+1$ onward tends to 0. Hence the tails of $\sum a_n b_n$ tend to 0. The Cauchy criterion proves convergence. □

Theorem 19.3 (Abel test for series). *If $\sum a_n$ converges and (b_n) is monotone and bounded, then $\sum a_n b_n$ converges.*

Proof. Since (b_n) is monotone and bounded, $b_n \to L$. Write

$$a_n b_n = a_n(b_n - L) + L a_n.$$

The series $\sum L a_n$ converges. Since $\sum a_n$ converges, its partial sums are bounded. The sequence $b_n - L$ is monotone and tends to 0. Dirichlet's test applies to $\sum a_n(b_n - L)$. Thus $\sum a_n b_n$ converges. □

Practice problems.

1. Use the integral test to determine convergence of $\sum 1/n^p$.
2. Prove convergence of $\sum \sin n/n$ using Dirichlet's test, assuming boundedness of the partial sums of $\sin n$.
3. Prove Abel's test from Dirichlet's test without using any integral language.

Ticket 20. Rearrangement of absolutely convergent series

Core idea. Absolute convergence means that the total mass of the tail is small, so rearranging terms cannot change the limit.

Definition 20.1 (Rearrangement). *A rearrangement of $\sum a_n$ is a series $\sum a_{\pi(n)}$, where $\pi : \mathbb{N} \to \mathbb{N}$ is a bijection.*

Theorem 20.1 (Rearrangement theorem for absolute convergence). *If $\sum a_n$ converges absolutely, then every rearrangement $\sum a_{\pi(n)}$ converges absolutely and*

$$\sum_{n=1}^{\infty} a_{\pi(n)} = \sum_{n=1}^{\infty} a_n.$$

Proof. Absolute convergence of the rearranged series follows because finite partial sums of $\sum |a_{\pi(n)}|$ are finite subsums of $\sum |a_n|$, hence bounded by $\sum |a_n|$. Therefore the rearranged nonnegative series converges.

Let $S = \sum a_n$. Given $\varepsilon > 0$, choose N such that

$$\sum_{n>N} |a_n| < \varepsilon.$$

Since π is a bijection, there exists M such that the set $\{1, 2, \ldots, N\}$ is contained in $\{\pi(1), \ldots, \pi(M)\}$. For $m \ge M$, the difference between the rearranged partial sum $\sum_{k=1}^{m} a_{\pi(k)}$ and S consists only of omitted and possibly extra terms with original index greater than N. Hence its absolute value is at most the tail $\sum_{n>N} |a_n| < \varepsilon$. Thus the rearranged partial sums converge to S. □

Practice problems.

1. Prove that finite rearrangements of any convergent series do not change its sum.
2. Give a detailed proof that $\sum |a_{\pi(n)}|$ converges if $\sum |a_n|$ converges.
3. State what can go wrong for conditionally convergent series and give the example of the alternating harmonic series.

Ticket 21. Products of absolutely convergent series

Core idea. The product of absolutely convergent series is justified by absolute convergence of the double series. The Cauchy product is a diagonal enumeration of that double series.

Definition 21.1 (Cauchy product)**.** *Given two series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, their Cauchy product is $\sum_{n=0}^{\infty} c_n$, where*

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Theorem 21.1 (Product theorem)**.** *If $\sum a_n$ and $\sum b_n$ converge absolutely, then the Cauchy product $\sum c_n$ converges absolutely and*

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right).$$

Proof. First prove absolute convergence. Since

$$|c_n| \le \sum_{k=0}^{n} |a_k||b_{n-k}|,$$

we have for partial sums

$$\sum_{n=0}^{N} |c_n| \le \sum_{n=0}^{N}\sum_{k=0}^{n} |a_k||b_{n-k}| \le \left(\sum_{k=0}^{N} |a_k|\right)\left(\sum_{j=0}^{N} |b_j|\right) \le \left(\sum_{k=0}^{\infty} |a_k|\right)\left(\sum_{j=0}^{\infty} |b_j|\right).$$

Thus $\sum |c_n|$ converges.

Let

$$A = \sum a_n, \qquad B = \sum b_n.$$

The finite product of partial sums is

$$\left(\sum_{k=0}^{N} a_k\right)\left(\sum_{j=0}^{N} b_j\right) = \sum_{0\le k,j\le N} a_k b_j.$$

As $N \to \infty$, this tends to AB. Absolute convergence of the double series $\sum_{k,j} |a_k b_j| = (\sum |a_k|)(\sum |b_j|)$ permits rearrangement of the double series by diagonals $k + j = n$. Therefore

$$AB = \sum_{k,j\ge 0} a_k b_j = \sum_{n=0}^{\infty}\sum_{k=0}^{n} a_k b_{n-k} = \sum_{n=0}^{\infty} c_n.$$

This proves the theorem. □

Practice problems.

1. Compute the Cauchy product of $\sum_{n=0}^{\infty} x^n$ with itself for $|x| < 1$.
2. Prove absolute convergence of the double series $\sum_{k,j} |a_k b_j|$ from absolute convergence of the two original series.
3. Explain why absolute convergence is the hypothesis that allows diagonal rearrangement.

Final synthesis

The central implication structure is as follows:

uniform continuity on compact intervals $\Longrightarrow$ small oscillation on fine partitions $\Longrightarrow$ Riemann integrability.

$$\text{absolute convergence} \Longrightarrow \text{Cauchy tails} \Longrightarrow \text{convergence}.$$

$$\text{bounded primitive} + \text{monotone factor to } 0 \Longrightarrow \text{Dirichlet convergence}.$$

absolute convergence of two series $\Longrightarrow$ absolute convergence of the double series $\Longrightarrow$ validity of Cauchy product.

Thus the course is not a list of isolated facts. It is a system: compactness gives uniformity; uniformity gives integrability; completeness gives Cauchy criteria; absolute convergence gives unconditional manipulation.

Appendix A. Detailed proof bank and structural expansions

This appendix is included to make the handout self-contained at the level needed for an oral exam. It records auxiliary facts that are often used silently in short lecture notes.

A.1. The mean value theorem package

Theorem 21.2 (Rolle theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$, differentiable on (a, b), and satisfy $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that $f'(c) = 0$.*

Proof. By the extreme value theorem, f attains a maximum and a minimum on $[a, b]$. If both extrema occur only at endpoints, then since $f(a) = f(b)$, the function is constant, and any $c \in (a, b)$ works. Otherwise, one extremum occurs at an interior point c. At an interior maximum or minimum, the derivative must be zero: for a maximum,

$$\frac{f(c+h) - f(c)}{h} \le 0 \quad (h > 0), \qquad \frac{f(c+h) - f(c)}{h} \ge 0 \quad (h < 0),$$

and the two one-sided limits force $f'(c) = 0$. The minimum case is analogous. □

Theorem 21.3 (Lagrange mean value theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$f(b) - f(a) = f'(c)(b - a).$$

Proof. Define the auxiliary function

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then g is continuous on $[a, b]$, differentiable on (a, b), and $g(a) = g(b)$. Rolle theorem gives $c \in (a, b)$ with $g'(c) = 0$. Therefore

$$f'(c) - \frac{f(b) - f(a)}{b - a} = 0,$$

which is the desired formula. □

A.2. Heine–Borel facts used in the integration theory

Theorem 21.4 (Extreme value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous, then f is bounded and attains its maximum and minimum.*

Proof. Since $[a, b]$ is compact and f is continuous, $f([a, b])$ is compact in $\mathbb{R}$. Compact subsets of $\mathbb{R}$ are closed and bounded. Boundedness gives finite $m = \inf f([a, b])$ and $M = \sup f([a, b])$. Since the image is closed, it contains m and M. Thus there exist $x_m, x_M \in [a, b]$ with $f(x_m) = m$ and $f(x_M) = M$. □

Theorem 21.5 (Intermediate value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.*

Proof. Assume $f(a) < y < f(b)$; the other cases are analogous. Let

$$E = \{x \in [a, b] : f(x) < y\}.$$

Then E is nonempty and bounded above, so $c = \sup E$ exists. Continuity at c rules out $f(c) < y$, because then points slightly to the right of c would still lie in E, contradicting the supremum property. It also rules out $f(c) > y$, because then points of E sufficiently close to c from the left could not exist, again contradicting the definition of supremum. Therefore $f(c) = y$. □

A.3. Full equivalence of Riemann and Darboux definitions

Theorem 21.6 (Riemann equals Darboux)**.** *For a bounded function $f : [a, b] \to \mathbb{R}$, the fine-tagged-sum definition of integrability is equivalent to Darboux integrability, and the two integrals are equal.*

Proof. Assume Darboux integrability and let I be the common Darboux integral. Given $\varepsilon > 0$, choose a partition P_0 such that $U(f, P_0) - L(f, P_0) < \varepsilon/3$. By the Darboux convergence theorem, if P is sufficiently fine, then its common refinement $R = P \cup P_0$ satisfies that both $U(f, P)$ and $L(f, P)$ differ from the refined estimates by less than $\varepsilon/3$. Any tagged sum over P lies between $L(f, P)$ and $U(f, P)$, hence lies within ε of I. Thus fine tagged sums converge to I.

Conversely, assume fine tagged sums converge to I. Given $\varepsilon > 0$, choose $\delta > 0$ such that every tagged sum over a partition of mesh less than δ is within $\varepsilon/4$ of I. Choose one such partition P. On each subinterval, select a tag whose value is within an arbitrarily small amount of the supremum, and another tag whose value is within an arbitrarily small amount of the infimum. The resulting upper-like and lower-like tagged sums differ by at most $\varepsilon/2$, up to an arbitrarily small error. Hence $U(f, P) - L(f, P) < \varepsilon$. The Darboux criterion follows. □

A.4. Why finite changes do not matter, but dense changes do

Changing a function at finitely many points does not change its Riemann integral because those points can be isolated in intervals of arbitrarily small total length. The same is not true for dense sets of points. For example, the function $1_{\mathbb{Q}}$ on $[0, 1]$ equals 1 on a dense set and 0 on a dense set. On every nondegenerate subinterval, its supremum is 1 and its infimum is 0. Hence every upper Darboux sum is 1, every lower Darboux sum is 0, and the function is not Riemann integrable.

A.5. The Cauchy philosophy

The same proof pattern appears in Riemann sums, improper integrals, and series.

Object	Partial object	Cauchy tail condition
Riemann integral	fine tagged sums	two fine sums are close
Improper integral	$F(A) = \int_a^A f$	$\int_C^B f$ is small for large B, C
Series	partial sums s_N	$\sum_{k=n+1}^m a_k$ is small
Uniform series	partial sums in $B(E)$	sup-norm tails are small

The reason this works is completeness: $\mathbb{R}$, $\mathbb{C}$, and complete normed spaces convert Cauchy conditions into genuine limits.

A.6. Abel summation as discrete integration by parts

Abel summation is the exact discrete analogue of integration by parts. If $A_n = \sum_{k=1}^n a_k$, then $a_k = A_k - A_{k-1}$. Thus

$$\sum_{k=p}^{q} a_k b_k = \sum_{k=p}^{q} (A_k - A_{k-1}) b_k.$$

Expanding and shifting the second sum gives boundary terms plus increments of b_k:

$$\sum_{k=p}^{q} a_k b_k = A_q b_q - A_{p-1} b_p + \sum_{k=p}^{q-1} A_k (b_k - b_{k+1}).$$

This is the discrete formula behind both Dirichlet and Abel tests. The boundedness of A_k replaces boundedness of a primitive in the integral version.

A.7. Product of series through absolute convergence of a double series

If $\sum |a_n| < \infty$ and $\sum |b_n| < \infty$, then

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} |a_i b_j| = \left(\sum_{i=0}^{\infty} |a_i| \right) \left(\sum_{j=0}^{\infty} |b_j| \right) < \infty.$$

This absolute convergence means that every enumeration of the pairs (i, j) gives the same sum. The Cauchy product is the enumeration by diagonals $i + j = n$. Therefore the product theorem for absolutely convergent series is not a formal algebraic trick; it is a consequence of unconditional convergence of an absolutely summable double family.

A.8. Minimal counterexample list

1. A continuous function on a noncompact interval need not be uniformly continuous: $f(x) = x^2$ on $\mathbb{R}$.
2. A bounded function need not be Riemann integrable: $1_{\mathbb{Q}}$ on $[0, 1]$.
3. An improper integral may converge conditionally but not absolutely: $\int_1^\infty \sin x/x \, dx$.
4. A series may converge conditionally but rearrangements may change its sum: the alternating harmonic series.
5. The ratio and root tests can be inconclusive: $\sum 1/n^2$ converges and $\sum 1/n$ diverges, but both have ratio and root limit equal to 1.

Analysis Tickets:

Topological and Filter-Theoretic Version with Practice Problems

Metric Spaces, Compactness, Uniform Convergence, Integrals, and Power Series

A rewritten lecture handout using neighborhood filters, ultrafilters, Cauchy filters, metric uniformities, and exam-oriented practice problems.

This handout rewrites a standard first analysis course in a more structural language. The order of ideas is:

metric spaces $\Rightarrow$ topologies $\Rightarrow$ neighborhood filters $\Rightarrow$ compactness via ultrafilters $\Rightarrow$ uniformities and Cauchy filters.

When a theorem is originally metric, the abstract formulation is immediately expanded back into the usual ε-δ or sequence statement.

Generated: June 16, 2026

Contents

How to use these notes 2

Conventions 2

1 Metric spaces as topological spaces 2

2 Closure, limit points, and convergence 4

3 Cauchy filters, completeness, and metric completeness 5

4 Compactness, ultrafilters, and closed-set principles 7

5 Total boundedness and compact metric spaces 8

6 Sequential compactness as a metric shadow of compactness 9

7 Limits and continuous mappings through filters 11

8 Connectedness and continuous images 12

9 Uniform continuity and Cantor theorem 14

10 Pointwise and uniform convergence of functions 15

11 Continuity of uniform limits 17

12 Integrals and derivatives as continuous operations 18

12.1 Differentiating uniform limits 19

13 Parameter integrals as maps into function spaces 20

14 Power series and local uniform convergence 21

15 Termwise integration and differentiation of power series 22

16 Analytic functions, Taylor coefficients, and elementary expansions 24

Final checklist 25

One-page dependency map 26

How to use these notes

A correct oral answer in analysis should contain four parts: the exact definitions, the theorem with all hypotheses, the proof mechanism, and a counterexample or warning showing why the hypotheses matter. The language of filters is useful because it makes several different-looking statements identical:

$$\text{continuity} = \text{preservation of convergent filters},$$

$$\text{compactness} = \text{convergence of ultrafilters},$$

$$\text{completeness} = \text{convergence of Cauchy filters},$$

$$\text{uniform continuity} = \text{preservation of entourages}.$$

The price is that one must distinguish carefully between topology and uniform structure. Ordinary continuity is topological; uniform continuity and Cauchy behavior are not purely topological.

Conventions

Unless stated otherwise, (X, d), (Y, ρ), and (Z, δ) are metric spaces. Their induced topologies are used silently. The scalar field $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. For a topological space X, $\mathcal{N}_X(x)$ denotes the neighborhood filter of x. For a map $f : X \to Y$, $f_*\mathcal{F}$ denotes the image filter.

Ticket 1. Metric spaces as topological spaces

Core idea. A metric gives numerical distances, but the first structure used by continuity is the topology generated by metric balls.

Definition 1.1 (Metric)**.** *A metric on a set X is a function $d : X \times X \to [0, \infty)$ such that, for all $x, y, z \in X$,*

$$d(x, y) = 0 \iff x = y, \qquad d(x, y) = d(y, x),$$
$$d(x, z) \le d(x, y) + d(y, z).$$

The pair (X, d) is a metric space.

Definition 1.2 (Metric topology)**.** *For $a \in X$ and $r > 0$, put*

$$B_d(a, r) = \{x \in X : d(x, a) < r\}.$$

A set $U \subset X$ is open if for every $x \in U$ there exists $r > 0$ such that $B_d(x, r) \subset U$. This defines the topology induced by d.

Proposition 1.1 (Topology axioms)**.** *The open sets induced by a metric satisfy:*

1. *$\varnothing$ and X are open;*
2. *arbitrary unions of open sets are open;*
3. *finite intersections of open sets are open.*

Proof. The first two statements are immediate. For finite intersections, let $x \in U_1 \cap \cdots \cap U_n$, where each U_i is open. Choose $r_i > 0$ such that $B(x, r_i) \subset U_i$. Then with $r = \min_i r_i$,

$$B(x, r) \subset U_1 \cap \cdots \cap U_n.$$

Thus the intersection is open. $\square$

Definition 1.3 (Neighborhood filter). *Let X be a topological space and $x \in X$. The neighborhood filter at x is*

$$\mathcal{N}_X(x) = \{A \subset X : \exists \text{ open } U \text{ with } x \in U \subset A\}.$$

In a metric space this is equivalently

$$\mathcal{N}_X(x) = \{A \subset X : \exists r > 0,\ B_d(x,r) \subset A\}.$$

Proposition 1.2. *For each $x \in X$, $\mathcal{N}_X(x)$ is a filter on X.*

Proof. The empty set is not a neighborhood of x. If $A, B \in \mathcal{N}_X(x)$, choose open sets U, V with $x \in U \subset A$ and $x \in V \subset B$. Then $U \cap V$ is open and

$$x \in U \cap V \subset A \cap B,$$

so $A \cap B \in \mathcal{N}_X(x)$. If $A \in \mathcal{N}_X(x)$ and $A \subset C \subset X$, then the same open neighborhood contained in A is contained in C, hence $C \in \mathcal{N}_X(x)$. □

Proposition 1.3 (Equivalent finite-dimensional metrics). *On $\mathbb{R}^n$, the metrics*

$$d_1(x,y) = \sum_{k=1}^{n} |x_k - y_k|, \quad d_2(x,y) = \left(\sum_{k=1}^{n} |x_k - y_k|^2\right)^{1/2}, \quad d_\infty(x,y) = \max_k |x_k - y_k|$$

generate the same topology and therefore the same neighborhood filters.

Proof. For all $x, y \in \mathbb{R}^n$,

$$d_\infty(x,y) \le d_2(x,y) \le d_1(x,y) \le n d_\infty(x,y).$$

Hence every ball for one metric contains a smaller ball for the others. Therefore the open sets and the neighborhood filters coincide. □

Common mistake. Same topology does not always mean same uniform structure. For example, on $\mathbb{R}$, the usual metric $d(x,y) = |x-y|$ and $\rho(x,y) = |\arctan x - \arctan y|$ generate the same topology, but the sequence n is ρ-Cauchy and not d-Cauchy.

Oral-exam minimum.

- Define the metric topology through balls.
- Define the neighborhood filter $\mathcal{N}_X(x)$.
- Prove that metric balls generate a topology.
- Explain why equivalent finite-dimensional norms give the same filters of neighborhoods.

Practice problems.

1. Prove directly that the discrete metric on a set X induces the discrete topology. Describe the neighborhood filter $\mathcal{N}_X(x)$.
2. Let d and ρ be metrics on X such that $cd(x,y) \le \rho(x,y) \le Cd(x,y)$ for some $c, C > 0$. Prove that they induce the same topology and the same neighborhood filters.
3. On $\mathbb{R}^n$, prove the inequalities $d_\infty \le d_2 \le d_1 \le n d_\infty$ and use them to compare balls explicitly.
4. Give an example of two metrics with the same open sets but different Cauchy sequences. Explain which structure changed and which did not.

Ticket 2. Closure, limit points, and convergence

Core idea. Closure means that every neighborhood sees the set. Filters remove the need to choose a sequence unless the space has a countable local base.

Definition 2.1 (Closure and interior)**.** *For $A \subset X$, the closure of A is*

$$\overline{A} = \bigcap \{F \subset X : F \text{ is closed and } A \subset F\}.$$

The interior is

$$\operatorname{Int} A = \{x \in A : A \in \mathcal{N}_X(x)\}.$$

The boundary is $\partial A = \overline{A} \setminus \operatorname{Int} A$.

Proposition 2.1 (Neighborhood criterion for closure)**.** *For a topological space X, $A \subset X$, and $x \in X$,*

$$x \in \overline{A} \iff \forall U \in \mathcal{N}_X(x),\ U \cap A \neq \varnothing.$$

Proof. If some neighborhood $U \in \mathcal{N}_X(x)$ is disjoint from A, choose an open O with $x \in O \subset U$. Then $X \setminus O$ is closed, contains A, and does not contain x. Hence $x \notin \overline{A}$.

Conversely, if $x \notin \overline{A}$, then $X \setminus \overline{A}$ is an open neighborhood of x disjoint from A. Thus not every neighborhood meets A. □

Definition 2.2 (Filter base attached to closure)**.** *If $x \in \overline{A}$, then*

$$\mathcal{B}_{A,x} = \{U \cap A : U \in \mathcal{N}_X(x)\}$$

is a filter base on A. The generated filter is the filter of points of A approaching x.

Why it is a filter base. Each $U \cap A$ is nonempty by the closure criterion. If $U, V \in \mathcal{N}_X(x)$, then $U \cap V \in \mathcal{N}_X(x)$, and

$$(U \cap V) \cap A \subset (U \cap A) \cap (V \cap A).$$

So the finite-intersection condition for a filter base holds. □

Definition 2.3 (Limit point)**.** *A point $x \in X$ is a limit point of $A \subset X$ if every neighborhood of x meets $A \setminus \{x\}$:*

$$\forall U \in \mathcal{N}_X(x), \quad U \cap (A \setminus \{x\}) \neq \varnothing.$$

Equivalently, $x \in \overline{A \setminus \{x\}}$.

Definition 2.4 (Sequence filter)**.** *A sequence (x_n) in X generates the filter*

$$\mathcal{F}_{(x_n)} = \{A \subset X : \exists N\ \forall n \geq N,\ x_n \in A\}.$$

Thus $A \in \mathcal{F}_{(x_n)}$ means that the sequence is eventually in A.

Definition 2.5 (Filter convergence)**.** *A filter $\mathcal{F}$ on a topological space X converges to x, written $\mathcal{F} \to x$, if*

$$\mathcal{N}_X(x) \subset \mathcal{F}.$$

For a sequence, this gives

$$x_n \to x \iff \mathcal{F}_{(x_n)} \to x.$$

Proposition 2.2 (Sequential closure in first-countable spaces)**.** *If X is first-countable, then*

$$x \in \overline{A} \iff \exists a_n \in A \text{ such that } a_n \to x.$$

In particular, this holds in every metric space.

Proof. If $a_n \in A$ and $a_n \to x$, then every neighborhood of x contains all sufficiently large a_n, hence meets A. So $x \in \overline{A}$.

Conversely, assume $x \in \overline{A}$. Since X is first-countable, choose a countable neighborhood base (U_n) at x. Replace it by the decreasing base

$$W_n = U_1 \cap \cdots \cap U_n.$$

Since $x \in \overline{A}$, each $W_n \cap A$ is nonempty. Choose $a_n \in W_n \cap A$. For any neighborhood O of x, some $W_N \subset O$. Since the W_n are decreasing, $a_n \in O$ for all $n \geq N$. Hence $a_n \to x$. ☐

Common mistake. Sequences do not characterize closure in arbitrary topological spaces. Filters do: $x \in \overline{A}$ exactly means that there exists a filter on A converging to x.

Oral-exam minimum.

- State closure via neighborhoods.
- Define sequence filters and filter convergence.
- Prove sequential closure under first-countability.
- Explain why filters are strictly more general than sequences.

Practice problems.

1. Prove that $x \in \overline{A}$ iff there exists a filter $\mathcal{F}$ on A such that $\mathcal{F} \to x$ in X.
2. In a metric space, prove that $x \in \overline{A}$ iff there is a sequence (a_n) in A with $a_n \to x$. Identify exactly where first countability is used.
3. Let $A \subset X$. Prove that $x \in \partial A$ iff every neighborhood of x meets both A and $X \setminus A$.
4. Construct a topological space where sequential closure is strictly smaller than topological closure. State why filters fix the problem.

Ticket 3. Cauchy filters, completeness, and metric completeness

Core idea. Cauchy behavior is not purely topological. It belongs to the uniform structure induced by the metric.

Definition 3.1 (Metric uniformity)**.** *For a metric space (X, d), define*

$$D^X_\varepsilon = \{(x, y) \in X \times X : d(x, y) < \varepsilon\}.$$

The metric uniformity $\mathcal{U}_X$ is the filter on $X \times X$ generated by the sets D^X_ε, $\varepsilon > 0$. Its elements are called entourages.

Definition 3.2 (Cauchy filter)**.** *A filter $\mathcal{F}$ on a metric space (X, d) is Cauchy if for every $\varepsilon > 0$ there exists $A \in \mathcal{F}$ such that*

$$\operatorname{diam}(A) < \varepsilon.$$

Equivalently,

$$\forall \varepsilon > 0\ \exists A \in \mathcal{F} : \quad A \times A \subset D^X_\varepsilon.$$

Proposition 3.1. *If $\mathcal{F} \to x$ in a metric space, then $\mathcal{F}$ is Cauchy.*

Proof. Let $\varepsilon > 0$. Since $\mathcal{F} \to x$, the ball $B(x, \varepsilon/2)$ belongs to $\mathcal{F}$. Its diameter is at most ε. Hence $\mathcal{F}$ is Cauchy. □

Definition 3.3 (Completeness through filters)**.** *A metric space X is complete if every Cauchy filter on X converges.*

Theorem 3.1 (Equivalence with sequential completeness)**.** *For metric spaces, the following are equivalent:*

1. *every Cauchy sequence converges;*
2. *every Cauchy filter converges.*

Proof. Assume every Cauchy filter converges. If (x_n) is a Cauchy sequence, its sequence filter $\mathcal{F}_{(x_n)}$ is Cauchy. Therefore $\mathcal{F}_{(x_n)} \to x$ for some x, which is exactly $x_n \to x$.

Conversely, assume every Cauchy sequence converges. Let $\mathcal{F}$ be a Cauchy filter. For each n, choose $A_n \in \mathcal{F}$ with $\operatorname{diam}(A_n) < 1/n$. Put

$$B_n = A_1 \cap \cdots \cap A_n \in \mathcal{F}.$$

Then $\operatorname{diam}(B_n) < 1/n$ and $B_{n+1} \subset B_n$. Choose $x_n \in B_n$. If $m, n \geq N$, then $x_m, x_n \in B_N$, so $d(x_m, x_n) < 1/N$. Hence (x_n) is Cauchy, and by assumption $x_n \to x$ for some $x \in X$.

We prove $\mathcal{F} \to x$. Let $\varepsilon > 0$. Choose N such that $1/N < \varepsilon/2$ and $d(x_N, x) < \varepsilon/2$. For every $y \in B_N$,

$$d(y, x) \leq d(y, x_N) + d(x_N, x) < \varepsilon.$$

Thus $B_N \subset B(x, \varepsilon)$. Since $B_N \in \mathcal{F}$, upward closure gives $B(x, \varepsilon) \in \mathcal{F}$. Therefore $\mathcal{F} \to x$. □

Proposition 3.2 (Closed subspaces of complete spaces)**.** *Let X be complete and $F \subset X$ closed. Then F is complete with the inherited metric. Conversely, if $F \subset X$ is complete with the inherited metric, then F is closed in X.*

Proof. If $\mathcal{F}$ is a Cauchy filter on F, then its upward extension to X is a Cauchy filter on X, hence converges to some $x \in X$. Since all members of the original filter live in F, the point x belongs to $\overline{F} = F$. Hence F is complete.

Conversely, suppose F is complete and $x \in \overline{F}$. The family $B(x, 1/n) \cap F$ generates a Cauchy filter on F, so it converges in F to some $y \in F$. But it also converges to x in X. Metric limits are unique, so $x = y \in F$. Hence F is closed. □

Oral-exam minimum.

- Define the metric uniformity $\mathcal{U}_X$.
- Define Cauchy filters and compare them with Cauchy sequences.
- Prove that, in metric spaces, sequence completeness and filter completeness coincide.
- Explain why Cauchy behavior is uniform, not merely topological.

Practice problems.

1. Let (x_n) be a sequence in a metric space. Prove that (x_n) is Cauchy iff its sequence filter is a Cauchy filter.
2. Prove that every convergent filter in a metric space is Cauchy.
3. Show that a complete subspace of a metric space is closed. Then prove the converse under the hypothesis that the ambient space is complete.
4. Prove that $\mathbb{R}^n$ is complete by using coordinate projections and the max metric.

Ticket 4. Compactness, ultrafilters, and closed-set principles

Core idea. Compactness says that no consistent system of closed constraints can escape the space. Ultrafilters express the same idea as convergence of maximal "eventually" structures.

Definition 4.1 (Compactness)**.** *A topological space X is compact if every open cover of X has a finite subcover.*

Definition 4.2 (Finite intersection property)**.** *A family $\{F_\alpha\}_{\alpha\in I}$ has the finite intersection property if every finite intersection is nonempty:*

$$F_{\alpha_1} \cap \cdots \cap F_{\alpha_n} \neq \varnothing.$$

Theorem 4.1 (Closed-set form of compactness)**.** *A topological space X is compact if and only if every family of closed subsets with the finite intersection property has nonempty total intersection.*

Proof. Let $F_\alpha = X \setminus U_\alpha$. The family $\{U_\alpha\}$ is an open cover with no finite subcover exactly when $\{F_\alpha\}$ is a family of closed sets with the finite intersection property and empty total intersection. Taking complements proves the equivalence. □

Definition 4.3 (Ultrafilter)**.** *A filter $\mathcal{U}$ on X is an ultrafilter if it is maximal among proper filters: there is no proper filter $\mathcal{G}$ with*

$$\mathcal{U} \subsetneq \mathcal{G}.$$

Equivalently, for every $A \subset X$, exactly one of the following holds:

$$A \in \mathcal{U}, \qquad X \setminus A \in \mathcal{U}.$$

Example 4.1 (Principal ultrafilter)*.* For $x \in X$,

$$\mathcal{U}_x = \{A \subset X : x \in A\}$$

is an ultrafilter. It converges to x in every topology on X.

Theorem 4.2 (Ultrafilter compactness criterion)**.** *A topological space X is compact if and only if every ultrafilter on X converges to at least one point of X.*

Proof. Assume X is compact, and let $\mathcal{U}$ be an ultrafilter on X. If $\mathcal{U}$ does not converge to any point, then for each $x \in X$ choose a neighborhood $U_x \in \mathcal{N}_X(x)$ such that $U_x \notin \mathcal{U}$. Since $\mathcal{U}$ is an ultrafilter, $X \setminus U_x \in \mathcal{U}$. The sets U_x cover X. By compactness, finitely many cover X:

$$X = U_{x_1} \cup \cdots \cup U_{x_n}.$$

Then

$$\varnothing = (X \setminus U_{x_1}) \cap \cdots \cap (X \setminus U_{x_n}) \in \mathcal{U},$$

a contradiction.

Conversely, suppose every ultrafilter converges. If X is not compact, then there is a family of closed sets $\{F_\alpha\}$ with the finite intersection property and empty intersection. The finite intersections of the F_α's form a filter base; extend the generated filter to an ultrafilter $\mathcal{U}$. By assumption $\mathcal{U} \to x$ for some $x \in X$. Since each $F_\alpha \in \mathcal{U}$ and F_α is closed, convergence forces $x \in F_\alpha$ for every α. Thus $x \in \bigcap_\alpha F_\alpha$, contradiction. □

Proposition 4.1 (Finite products of compact spaces)**.** *If X and Y are compact, then $X \times Y$ is compact.*

Proof. Let $\mathcal{U}$ be an ultrafilter on $X \times Y$. The projection image filters $(\pi_X)_*\mathcal{U}$ and $(\pi_Y)_*\mathcal{U}$ extend to ultrafilters and therefore converge to some $x \in X$ and $y \in Y$. For a basic neighborhood $O \times V$ of (x, y), the sets $\pi_X^{-1}(O)$ and $\pi_Y^{-1}(V)$ belong to $\mathcal{U}$, so their intersection $O \times V$ belongs to $\mathcal{U}$. Hence $\mathcal{U} \to (x, y)$. By the ultrafilter criterion, $X \times Y$ is compact. □

Oral-exam minimum.

- Prove the closed-set compactness criterion.
- Define ultrafilter and principal ultrafilter.
- Prove compactness iff every ultrafilter converges.
- Use ultrafilters to prove finite products of compact spaces are compact.

Practice problems.

1. Prove that an ultrafilter $\mathcal{U}$ has the dichotomy property: for every $A \subset X$, exactly one of A and $X \setminus A$ belongs to $\mathcal{U}$.
2. Prove that the closed-set finite intersection property is equivalent to open-cover compactness.
3. Prove that a topological space is compact iff every ultrafilter on it converges.
4. Let $f : X \to Y$ be continuous and X compact. Prove $f(X)$ is compact using ultrafilters only.

Ticket 5. Total boundedness and compact metric spaces

Core idea. In metric spaces, compactness is equivalent to completeness plus total boundedness. Filters give a concise proof: every ultrafilter becomes Cauchy and then converges.

Definition 5.1 (Total boundedness)**.** *A metric space (X, d) is totally bounded if for every $\varepsilon > 0$ there exist finitely many points $x_1, \dots, x_N \in X$ such that*

$$X \subset \bigcup_{j=1}^{N} B(x_j, \varepsilon).$$

For a subset $A \subset X$, this is applied to the inherited metric on A.

Proposition 5.1. *Every totally bounded metric space is bounded. The converse is false in general.*

Proof. If $X \subset \bigcup_{j=1}^{N} B(x_j, 1)$, then all of X lies in a sufficiently large ball around x_1, because the finite set $\{x_1, \dots, x_N\}$ is bounded. The converse fails in infinite-dimensional normed spaces: in ℓ^∞, the unit vectors are bounded but pairwise distance 1, so no finite 1/3-net exists. □

Lemma 5.1 (Ultrafilters in totally bounded spaces are Cauchy)**.** *If X is totally bounded, then every ultrafilter on X is Cauchy.*

Proof. Let $\mathcal{U}$ be an ultrafilter and let $\varepsilon > 0$. Choose a finite cover

$$X \subset B(x_1, \varepsilon/2) \cup \cdots \cup B(x_N, \varepsilon/2).$$

Since $\mathcal{U}$ is an ultrafilter, one of these balls belongs to $\mathcal{U}$; otherwise all their complements would belong to $\mathcal{U}$, and their finite intersection would be empty. That ball has diameter at most ε. Hence $\mathcal{U}$ is Cauchy. □

Theorem 5.1 (Compact metric criterion)**.** *For a metric space X,*

$$X \text{ compact} \iff X \text{ complete and totally bounded.}$$

Proof. Assume X is compact. To prove total boundedness, suppose not. Then for some $\varepsilon > 0$, no finite ε-net exists. Inductively choose x_{n+1} outside $B(x_1, \varepsilon) \cup \cdots \cup B(x_n, \varepsilon)$. Then $d(x_m, x_n) \geq \varepsilon$ for $m \neq n$. The sequence filter can be extended to an ultrafilter $\mathcal{U}$. Compactness gives $\mathcal{U} \to x$, hence $\mathcal{U}$ is Cauchy. But the separated construction prevents any set in the tail filter from having diameter $< \varepsilon$, contradiction. Thus X is totally bounded. Completeness follows because every Cauchy filter extends to an ultrafilter, and compactness makes that ultrafilter converge; the original Cauchy filter, being coarser than a convergent filter and Cauchy, converges to the same limit.

Conversely, assume X is complete and totally bounded. Let $\mathcal{U}$ be an ultrafilter on X. By total boundedness, $\mathcal{U}$ is Cauchy. By completeness, $\mathcal{U}$ converges. Hence every ultrafilter converges, so X is compact. □

Corollary 5.1 (Heine–Borel in $\mathbb{R}^n$)**.** *A subset $A \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. A closed subset of the complete space $\mathbb{R}^n$ is complete. A bounded subset of $\mathbb{R}^n$ is totally bounded by subdivision into finitely many small cubes. Hence closed and bounded implies compact. Conversely, compact metric spaces are complete and totally bounded; in particular they are bounded and closed in any ambient metric space. □

Oral-exam minimum.

- Define total boundedness.
- Prove total boundedness implies boundedness and give an infinite-dimensional counterexample to the converse.
- Prove complete plus totally bounded implies compact using ultrafilters.
- Deduce Heine–Borel in $\mathbb{R}^n$.

Practice problems.

1. Prove that total boundedness implies boundedness in every metric space.
2. Give a bounded subset of an infinite-dimensional normed space that is not totally bounded, and prove it.
3. Prove that every bounded subset of $\mathbb{R}^n$ is totally bounded.
4. Prove that a compact metric space is totally bounded using an open cover by ε-balls.

Ticket 6. Sequential compactness as a metric shadow of compactness

Core idea. In metric spaces, compactness and sequential compactness coincide because first-countability lets filters produce sequences and total boundedness lets sequences produce Cauchy subsequences.

Definition 6.1 (Sequential compactness)**.** *A metric space K is sequentially compact if every sequence in K has a convergent subsequence whose limit belongs to K.*

Theorem 6.1 (Compact metric implies sequentially compact)**.** *Every compact metric space is sequentially compact.*

Proof. Let (x_n) be a sequence in compact K. Its sequence filter $\mathcal{F}_{(x_n)}$ extends to an ultrafilter $\mathcal{U}$. Since K is compact, $\mathcal{U} \to x$ for some $x \in K$.

We construct a subsequence converging to x. Since $\mathcal{U} \to x$, each ball $B(x, 1/k)$ belongs to $\mathcal{U}$. Each tail $\{x_n : n \geq N\}$ also belongs to $\mathcal{U}$. Therefore their intersection is nonempty. Choose recursively

$$n_k > n_{k-1}, \qquad x_{n_k} \in B(x, 1/k).$$

Then $x_{n_k} \to x$. Hence K is sequentially compact. □

Theorem 6.2 (Sequentially compact metric implies compact)**.** *Every sequentially compact metric space is compact.*

Proof. First, K is totally bounded. If not, for some $\varepsilon > 0$ one can choose a sequence (x_n) with $d(x_m, x_n) \geq \varepsilon$ for $m \neq n$. Such a sequence has no Cauchy subsequence and hence no convergent subsequence, contradiction.

Second, K is complete. Let (x_n) be Cauchy. By sequential compactness it has a subsequence (x_{n_j}) converging to $x \in K$. Since the whole sequence is Cauchy, the whole sequence converges to the same x. Therefore K is complete.

By the compact metric criterion, complete plus totally bounded implies compact. □

Corollary 6.1. *For metric spaces,*

$$\boxed{\textit{compact} \iff \textit{sequentially compact} \iff \textit{complete and totally bounded.}}$$

Lemma 6.1 (Lebesgue number lemma)**.** *Let K be compact metric, and let $\{U_\alpha\}$ be an open cover of K. Then there exists $\lambda > 0$ such that every subset of K with diameter less than λ is contained in some U_α.*

Proof. Suppose not. For each n, choose $A_n \subset K$ with $\operatorname{diam}(A_n) < 1/n$ such that A_n is contained in no member of the cover. Pick $x_n \in A_n$. By compactness, pass to a subsequence $x_{n_j} \to x \in K$. Choose α and $r > 0$ such that $B(x, r) \subset U_\alpha$. For large j, $d(x_{n_j}, x) < r/2$ and $1/n_j < r/2$. If $y \in A_{n_j}$, then

$$d(y, x) \leq d(y, x_{n_j}) + d(x_{n_j}, x) < r.$$

Thus $A_{n_j} \subset U_\alpha$, contradiction. □

Oral-exam minimum.

- Prove compact metric implies sequentially compact using ultrafilters.
- Prove sequentially compact metric implies complete and totally bounded.
- State the compact metric equivalence triangle.
- Prove the Lebesgue number lemma.

Practice problems.

1. Prove that compact metric spaces are sequentially compact using cluster points.
2. Prove that sequentially compact metric spaces are totally bounded.
3. Prove the Lebesgue number lemma for sequentially compact metric spaces.
4. Combine the previous results to prove the metric equivalence: compact iff sequentially compact iff complete and totally bounded.

Ticket 7. Limits and continuous mappings through filters

Core idea. A function is continuous if it sends the neighborhood filter of a point to a filter converging to the image point.

Definition 7.1 (Image filter)**.** *Let $f : X \to Y$, and let $\mathcal{F}$ be a filter on X. The image filter is*

$$f_*\mathcal{F} = \{B \subset Y : f^{-1}(B) \in \mathcal{F}\}.$$

Definition 7.2 (Limit along a filter)**.** *Let Y be topological. We say*

$$\lim_{\mathcal{F}} f = b$$

if

$$f_*\mathcal{F} \to b,$$

or equivalently,

$$\forall V \in \mathcal{N}_Y(b), \qquad f^{-1}(V) \in \mathcal{F}.$$

Definition 7.3 (Punctured limit along a subset)**.** *Let $A \subset X$, $a \in \overline{A}$, and $f : A \to Y$. The punctured neighborhood filter of a along A is*

$$\mathcal{N}_A^\circ(a) = \{U \subset A : \exists W \in \mathcal{N}_X(a),\ (W \cap A) \setminus \{a\} \subset U\}.$$

Then

$$\lim_{x \to a,\ x \in A,\ x \neq a} f(x) = b$$

means

$$\lim_{\mathcal{N}_A^\circ(a)} f = b.$$

In metric spaces this is exactly

$$\forall \varepsilon > 0\ \exists \delta > 0 : \quad x \in A,\ 0 < d(x, a) < \delta \Rightarrow \rho(f(x), b) < \varepsilon.$$

Theorem 7.1 (Continuity at a point)**.** *For $f : X \to Y$,*

$$f \text{ is continuous at } x \quad \Longleftrightarrow \quad f_*\mathcal{N}_X(x) \to f(x).$$

Equivalently,

$$\mathcal{N}_Y(f(x)) \subset f_*\mathcal{N}_X(x).$$

Proof. The condition $f_*\mathcal{N}_X(x) \to f(x)$ means

$$\forall V \in \mathcal{N}_Y(f(x)), \quad V \in f_*\mathcal{N}_X(x).$$

By definition of image filter,

$$V \in f_*\mathcal{N}_X(x) \quad \Longleftrightarrow \quad f^{-1}(V) \in \mathcal{N}_X(x).$$

Thus the condition becomes

$$\forall V \in \mathcal{N}_Y(f(x)), \quad f^{-1}(V) \in \mathcal{N}_X(x),$$

which is exactly the topological definition of continuity at x. □

Theorem 7.2 (Continuity preserves all convergent filters)**.** *A map $f : X \to Y$ is continuous if and only if for every filter $\mathcal{F}$ on X,*

$$\mathcal{F} \to x \quad \Longrightarrow \quad f_*\mathcal{F} \to f(x).$$

Proof. If f is continuous and $\mathcal{F} \to x$, let $V \in \mathcal{N}_Y(f(x))$. Then $f^{-1}(V) \in \mathcal{N}_X(x)$. Since $\mathcal{N}_X(x) \subset \mathcal{F}$, we get $f^{-1}(V) \in \mathcal{F}$. Hence $V \in f_*\mathcal{F}$, so $f_*\mathcal{F} \to f(x)$.

Conversely, apply the stated property to $\mathcal{F} = \mathcal{N}_X(x)$. Then $f_*\mathcal{N}_X(x) \to f(x)$, so f is continuous at x. Since x is arbitrary, f is continuous. ☐

Theorem 7.3 (Heine criterion in first-countable spaces)**.** *If X is first-countable, then $f : X \to Y$ is continuous at x if and only if*

$$x_n \to x \quad \Longrightarrow \quad f(x_n) \to f(x)$$

for every sequence (x_n) in X.

Proof. Continuity implies the sequential statement because sequence filters are filters.

Conversely, assume the sequential statement holds but f is not continuous at x. Then for some $V \in \mathcal{N}_Y(f(x))$, the preimage $f^{-1}(V)$ is not a neighborhood of x. Let (W_n) be a decreasing countable local base at x. For each n, choose

$$x_n \in W_n \setminus f^{-1}(V).$$

Then $x_n \to x$, but $f(x_n) \notin V$ for all n, so $f(x_n) \not\to f(x)$. Contradiction. ☐

Oral-exam minimum.

- Define image filters.
- Write continuity at x as $f_*\mathcal{N}_X(x) \to f(x)$.
- Prove the preservation-of-filters criterion.
- Explain why the sequence criterion needs first-countability.

Practice problems.

1. Let $f : X \to Y$. Prove that f is continuous at x iff $f_*\mathcal{N}_X(x) \to f(x)$.

2. Prove that f is continuous iff for every filter $\mathcal{F} \to x$, one has $f_*\mathcal{F} \to f(x)$.

3. In metric spaces, expand the filter definition of $\lim_{x\to a} f(x) = b$ into the usual punctured ε-δ definition.

4. Prove that in first-countable spaces sequential continuity implies continuity. Then give a reason why this can fail in general topological spaces.

Ticket 8. Connectedness and continuous images

Core idea. Connectedness is purely topological: it is exactly the impossibility of separating the space into two nonempty open pieces.

Definition 8.1 (Separation and connectedness)**.** *A topological space X is disconnected if there exist nonempty disjoint open sets $U, V \subset X$ such that*

$$X = U \cup V.$$

If no such pair exists, X is connected. A subset $A \subset X$ is connected if it is connected with the subspace topology.

Theorem 8.1 (Continuous images preserve connectedness)**.** *If X is connected and $f : X \to Y$ is continuous, then $f(X)$ is connected.*

Proof. Suppose $f(X) = A \cup B$ is a separation in the subspace topology of $f(X)$. Then

$$f^{-1}(A), \qquad f^{-1}(B)$$

are disjoint nonempty open subsets of X whose union is X. This contradicts connectedness of X. □

Theorem 8.2 (Intervals are connected)**.** *Every interval $I \subset \mathbb{R}$ is connected.*

Proof. Assume $I = A \cup B$ is a separation. Choose $a \in A$, $b \in B$, with $a < b$. Let

$$c = \sup(A \cap [a, b]).$$

Since I is an interval, $c \in I$. If $c \in A$, relative openness of A gives points of A to the right of c, contradicting the definition of supremum. If $c \in B$, relative openness of B gives a left neighborhood of c contained in B, contradicting that c is the supremum of points of $A \cap [a, b]$. Hence no separation exists. □

Theorem 8.3 (Intermediate value theorem)**.** *If K is connected and $f : K \to \mathbb{R}$ is continuous, then $f(K)$ is an interval. If additionally K is compact, then*

$$f(K) = [m, M],$$

where $m = \min_K f$ and $M = \max_K f$.

Proof. The image $f(K)$ is connected, hence an interval in $\mathbb{R}$. If K is compact, then $f(K)$ is compact, hence closed and bounded. Therefore it contains its infimum and supremum. □

Oral-exam minimum.

- Define connectedness through separations.
- Prove continuous images preserve connectedness.
- Prove intervals in $\mathbb{R}$ are connected.
- Derive the intermediate value theorem.

Practice problems.

1. Prove that X is connected iff every continuous map $X \to \{0, 1\}$, where $\{0, 1\}$ has the discrete topology, is constant.
2. Prove that the continuous image of a connected space is connected.
3. Prove that every interval in $\mathbb{R}$ is connected using the supremum argument.
4. Let K be compact and connected, and $f : K \to \mathbb{R}$ continuous. Prove that $f(K) = [m, M]$ for some $m, M \in \mathbb{R}$.

Ticket 9. Uniform continuity and Cantor theorem

Core idea. Uniform continuity is not pointwise. It says that closeness to the diagonal in $X \times X$ is sent to closeness to the diagonal in $Y \times Y$.

Definition 9.1 (Uniform continuity)**.** *For metric spaces (X, d) and (Y, ρ), a map $f : X \to Y$ is uniformly continuous if*

$$\forall \varepsilon > 0\ \exists \delta > 0 : \quad d(x, y) < \delta \Rightarrow \rho(f(x), f(y)) < \varepsilon.$$

Theorem 9.1 (Filter/uniformity form of uniform continuity)**.** *Let $\mathcal{U}_X$ and $\mathcal{U}_Y$ be the metric uniformities. Then $f : X \to Y$ is uniformly continuous if and only if*

$$\forall V \in \mathcal{U}_Y, \qquad (f \times f)^{-1}(V) \in \mathcal{U}_X.$$

Equivalently,

$$\mathcal{U}_Y \subset (f \times f)_* \mathcal{U}_X.$$

Proof. It is enough to test the basic entourages

$$D^Y_\varepsilon = \{(u, v) : \rho(u, v) < \varepsilon\}.$$

The condition

$$(f \times f)^{-1}(D^Y_\varepsilon) \in \mathcal{U}_X$$

means that for some $\delta > 0$,

$$D^X_\delta \subset (f \times f)^{-1}(D^Y_\varepsilon).$$

Expanding this inclusion gives exactly

$$d(x, y) < \delta \Rightarrow \rho(f(x), f(y)) < \varepsilon.$$

□

Theorem 9.2 (Cantor theorem / Heine–Cantor)**.** *Let K be a compact metric space and let Y be a metric space. If $f : K \to Y$ is continuous, then f is uniformly continuous.*

Filter proof. Assume f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$, there are $x, y \in K$ satisfying

$$d(x, y) < \delta, \qquad \rho(f(x), f(y)) \geq \varepsilon_0.$$

Let

$$D^K_\delta = \{(x, y) \in K \times K : d(x, y) < \delta\}$$

and define the bad set

$$C = \{(x, y) \in K \times K : \rho(f(x), f(y)) \geq \varepsilon_0\}.$$

The failure of uniform continuity says

$$D^K_\delta \cap C \neq \varnothing \quad \text{for every } \delta > 0.$$

Thus the family $\{D^K_\delta \cap C : \delta > 0\}$ is a filter base on $K \times K$. Let $\mathcal{F}$ be the generated filter. Then

$$C \in \mathcal{F}, \qquad D^K_\delta \in \mathcal{F} \quad (\delta > 0).$$

Extend $\mathcal{F}$ to an ultrafilter $\mathcal{G}$ on $K \times K$. Since K is compact, $K \times K$ is compact, so $\mathcal{G} \to (a, b)$ for some $(a, b) \in K \times K$.

Because $D^K_\delta \in \mathcal{G}$ for every $\delta > 0$, the limit must lie on the diagonal. Indeed, if $a \neq b$, choose $\delta < d(a,b)/3$. A sufficiently small product neighborhood of (a,b) is disjoint from D^K_δ, contradicting the fact that both sets belong to $\mathcal{G}$. Hence $a = b$, so $\mathcal{G} \to (a,a)$.

Continuity of f implies continuity of $f \times f$. Therefore

$$(f \times f)_*\mathcal{G} \to (f(a), f(a)).$$

The set

$$D^Y_{\varepsilon_0} = \{(u,v) : \rho(u,v) < \varepsilon_0\}$$

is a neighborhood of $(f(a), f(a))$, hence

$$(f \times f)^{-1}(D^Y_{\varepsilon_0}) \in \mathcal{G}.$$

But by definition of C,

$$C = (K \times K) \setminus (f \times f)^{-1}(D^Y_{\varepsilon_0}).$$

Since also $C \in \mathcal{G}$, the filter $\mathcal{G}$ contains two disjoint sets. Hence it contains $\varnothing$, impossible. Therefore f is uniformly continuous. □

Oral-exam minimum.

- State uniform continuity through $(f \times f)^{-1}$.
- Explain why uniform continuity requires a uniformity, not just a topology.
- Prove Cantor theorem using ultrafilters on $K \times K$.
- Identify exactly where compactness is used.

Practice problems.

1. Define the metric uniformity $\mathcal{U}_X$ and prove that it is a filter on $X \times X$.
2. Prove that $f : X \to Y$ is uniformly continuous iff $(f \times f)^{-1}(V) \in \mathcal{U}_X$ for every $V \in \mathcal{U}_Y$.
3. Prove Cantor's theorem using ultrafilters: if K is compact metric and $f : K \to Y$ is continuous, then f is uniformly continuous.
4. Give an example of a continuous function on a noncompact metric space that is not uniformly continuous. Identify where the compactness proof fails.

Ticket 10. Pointwise and uniform convergence of functions

Core idea. Pointwise convergence is convergence after every evaluation map. Uniform convergence is convergence in a function-space uniformity.

Definition 10.1 (Pointwise convergence)**.** *Let E be a set and (Y, ρ) a metric space. A sequence $f_n : E \to Y$ converges pointwise to $f : E \to Y$ if*

$$\forall x \in E, \qquad f_n(x) \to f(x).$$

Equivalently, for every evaluation map

$$\mathrm{ev}_x : Y^E \to Y, \qquad \mathrm{ev}_x(g) = g(x),$$

we have $\mathrm{ev}_x(f_n) \to \mathrm{ev}_x(f)$.

Definition 10.2 (Uniform convergence)**.** *If $Y = \mathbb{K}$ or more generally if Y is metric, $f_n : E \to Y$ converges uniformly to f if*

$$\forall \varepsilon > 0\ \exists N\ \forall n \geq N\ \forall x \in E : \rho(f_n(x), f(x)) < \varepsilon.$$

For scalar bounded functions this is convergence in the sup metric

$$d_\infty(f, g) = \sup_{x \in E} |f(x) - g(x)|.$$

Theorem 10.1 (Uniform Cauchy criterion)**.** *Let Y be complete. A sequence $f_n : E \to Y$ converges uniformly to some $f : E \to Y$ if and only if*

$$\forall \varepsilon > 0\ \exists N\ \forall m, n \geq N\ \forall x \in E : \rho(f_m(x), f_n(x)) < \varepsilon.$$

Proof. Uniform convergence implies the criterion by the triangle inequality. Conversely, the criterion makes $(f_n(x))$ Cauchy in Y for every fixed x. Since Y is complete, define

$$f(x) = \lim_n f_n(x).$$

Fix $\varepsilon > 0$. Choose N such that $\rho(f_m(x), f_n(x)) < \varepsilon$ for all $m, n \geq N$ and all x. Letting $m \to \infty$ gives

$$\rho(f(x), f_n(x)) \leq \varepsilon$$

for every x and all $n \geq N$. Hence $f_n \to f$ uniformly. □

Definition 10.3 (Uniform convergence of series)**.** *A series $\sum u_n(x)$ converges uniformly if its partial sums*

$$S_N(x) = \sum_{n=1}^{N} u_n(x)$$

converge uniformly.

Theorem 10.2 (Weierstrass M-test)**.** *If $|u_n(x)| \leq M_n$ for all $x \in E$ and $\sum M_n < \infty$, then $\sum u_n$ converges absolutely and uniformly on E.*

Proof. For $m > n$,

$$\sup_{x \in E} \left| \sum_{k=n+1}^{m} u_k(x) \right| \leq \sum_{k=n+1}^{m} M_k.$$

The right side tends to 0 as $m, n \to \infty$. The uniform Cauchy criterion applies. □

Oral-exam minimum.

- Distinguish pointwise from uniform convergence.
- Express pointwise convergence using evaluation maps.
- Prove the uniform Cauchy criterion.
- Prove the Weierstrass M-test.

Practice problems.

1. For $f_n : E \to \mathbb{K}$, prove that uniform convergence is exactly convergence in the sup metric whenever the suprema are finite.
2. Prove the uniform Cauchy criterion for sequences of functions $f_n : E \to \mathbb{K}$.
3. Show that $f_n(x) = x^n$ on $[0, 1]$ converges pointwise but not uniformly.
4. Prove the Weierstrass M-test using the uniform Cauchy criterion.

Ticket 11. Continuity of uniform limits

Core idea. Uniform convergence lets one control the limit function globally, while continuity controls one finite approximating function locally.

Theorem 11.1 (Uniform limit theorem)**.** *Let X be a topological space and (Y, ρ) a metric space. If $f_n : X \to Y$ are continuous and $f_n \to f$ uniformly, then f is continuous.*

Proof. Fix $x_0 \in X$ and $\varepsilon > 0$. Choose N such that

$$\rho(f_N(x), f(x)) < \varepsilon/3$$

for all $x \in X$. By continuity of f_N at x_0, the set

$$U = f_N^{-1}(B(f_N(x_0), \varepsilon/3))$$

is a neighborhood of x_0. If $x \in U$, then

$$\rho(f(x), f(x_0)) \leq \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(x_0)) + \rho(f_N(x_0), f(x_0)) < \varepsilon.$$

Thus $f^{-1}(B(f(x_0), \varepsilon)) \in \mathcal{N}_X(x_0)$, so f is continuous at x_0. □

Filter formulation. Let $\mathcal{F} \to x_0$. We must prove $f_*\mathcal{F} \to f(x_0)$. Given $\varepsilon > 0$, choose N as above. Since f_N is continuous,

$$(f_N)_*\mathcal{F} \to f_N(x_0).$$

Hence

$$f_N^{-1}(B(f_N(x_0), \varepsilon/3)) \in \mathcal{F}.$$

On this set, the same triangle estimate gives $f(x) \in B(f(x_0), \varepsilon)$. Therefore $f^{-1}(B(f(x_0), \varepsilon)) \in \mathcal{F}$, proving $f_*\mathcal{F} \to f(x_0)$. □

Corollary 11.1. *If $u_n : X \to \mathbb{K}$ are continuous and $\sum u_n$ converges uniformly, then its sum is continuous.*

Oral-exam minimum.

- Prove the uniform limit theorem with the $\varepsilon/3$ estimate.
- Rewrite the proof using convergent filters.
- Explain why pointwise convergence is not enough.

Practice problems.

1. Prove that if $f_n : X \to Y$ are continuous, Y is metric, and $f_n \to f$ uniformly, then f is continuous.
2. Rewrite the proof of uniform-limit continuity using neighborhood filters instead of $\varepsilon/3$ notation.
3. Give an example of pointwise convergence of continuous functions to a discontinuous function.
4. Prove that if $\sum u_n$ converges uniformly and every u_n is continuous, then the sum is continuous.

Ticket 12. Integrals and derivatives as continuous operations

Core idea. A limit passes through an integral because the integral is a continuous linear functional on the normed space $C([a,b])$.

Definition 12.1 (Normed vector space)**.** *A normed vector space is a vector space V over $\mathbb{K}$ with a norm $\|\cdot\|$. A linear map $L : V \to \mathbb{K}$ is bounded if there exists $C \geq 0$ such that*

$$|L(v)| \leq C\|v\| \quad (v \in V).$$

Proposition 12.1. *Every bounded linear functional is continuous.*

Proof. If $v_i \to v$, then

$$|L(v_i) - L(v)| = |L(v_i - v)| \leq C\|v_i - v\| \to 0.$$

In filter language, if $\mathcal{F} \to v$, then for every ball around $L(v)$, the inverse image contains a ball around v, hence belongs to $\mathcal{F}$. Thus $L_*\mathcal{F} \to L(v)$. □

Proposition 12.2 (Integral functional)**.** *On $C([a,b])$ with the uniform norm*

$$\|f\|_\infty = \sup_{x\in[a,b]} |f(x)|,$$

define

$$I(f) = \int_a^b f(x)\,dx.$$

Then I is a bounded linear functional and

$$|I(f)| \leq (b-a)\|f\|_\infty.$$

Proof. Linearity is the linearity of the Riemann integral. The estimate follows from

$$\left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx \leq (b-a)\|f\|_\infty.$$

□

Theorem 12.1 (Limit under the integral sign)**.** *If $f_n \in C([a,b])$ and $f_n \to f$ uniformly, then*

$$\int_a^b f_n(x)\,dx \to \int_a^b f(x)\,dx.$$

Proof. Uniform convergence is exactly $\|f_n - f\|_\infty \to 0$. Hence

$$\left|\int_a^b f_n - \int_a^b f\right| = |I(f_n - f)| \leq (b-a)\|f_n - f\|_\infty \to 0.$$

Equivalently, this is simply continuity of the map $I : C([a,b]) \to \mathbb{K}$. □

Corollary 12.1 (Termwise integration)**.** *If $u_n \in C([a,b])$ and $\sum u_n$ converges uniformly to S, then*

$$\int_a^b S(x)\,dx = \sum_{n=1}^{\infty} \int_a^b u_n(x)\,dx.$$

Oral-exam minimum.

- Define bounded linear functional.
- Prove bounded linear functionals are continuous.
- Prove the integral estimate $|I(f)| \leq (b-a)\|f\|_\infty$.
- Derive passage of limits through integrals.

12.1 Differentiating uniform limits

Core idea. Uniform convergence of derivatives plus convergence at one point reconstructs the functions by the fundamental theorem of calculus.

Theorem 12.2 (Differentiation theorem for uniform limits)**.** *Let $f_n \in C^1([a,b])$. Suppose $f_n(x_0)$ converges for some $x_0 \in [a,b]$, and $f_n' \to g$ uniformly on $[a,b]$. Then f_n converges uniformly to a function $f \in C^1([a,b])$, and*

$$f' = g.$$

Proof. By the fundamental theorem of calculus,

$$f_n(x) = f_n(x_0) + \int_{x_0}^{x} f_n'(t)\,dt.$$

Let $c = \lim_n f_n(x_0)$. Since $f_n' \to g$ uniformly,

$$\sup_{x\in[a,b]} \left| \int_{x_0}^{x} (f_n'(t) - g(t))\,dt \right| \le (b-a)\|f_n' - g\|_\infty \to 0.$$

Thus $f_n \to f$ uniformly, where

$$f(x) = c + \int_{x_0}^{x} g(t)\,dt.$$

Since g is continuous as a uniform limit of continuous functions, the fundamental theorem gives $f' = g$. □

Corollary 12.2 (Termwise differentiation of series)**.** *Let $u_n \in C^1([a,b])$. If $\sum u_n(x_0)$ converges at one point and $\sum u_n'$ converges uniformly on $[a,b]$, then $\sum u_n$ converges uniformly to a differentiable function S, and*

$$S'(x) = \sum_{n=1}^{\infty} u_n'(x).$$

Oral-exam minimum.

- State the hypotheses: convergence at one point plus uniform convergence of derivatives.
- Prove uniform convergence of f_n using the integral formula.
- Explain why pointwise convergence of derivatives is not enough.

Practice problems.

1. Prove that $I : C([a,b]) \to \mathbb{K}$, $I(f) = \int_a^b f(x)\,dx$, is a bounded linear functional and compute a valid operator norm bound.
2. Use the continuity of I to prove that uniform convergence permits passage of limits through integrals.
3. Prove termwise integration of a uniformly convergent series of continuous functions.
4. Prove the theorem on differentiating a uniformly convergent sequence under the hypotheses: $f_n(x_0)$ converges and $f_n' \to g$ uniformly.

Ticket 13. Parameter integrals as maps into function spaces

Core idea. A parameter integral is a composition: the parameter selects a function, and a continuous linear functional integrates it.

Definition 13.1. *Let T be a topological space. A map*

$$G : T \to C([a, b])$$

is continuous at t_0 if

$$G_*\mathcal{N}_T(t_0) \to G(t_0)$$

in the uniform norm topology. If $G(t)(x) = f(x, t)$, this means uniform convergence in x as $t \to t_0$:

$$\sup_{x\in[a,b]} |f(x, t) - f(x, t_0)| \to 0.$$

Theorem 13.1 (Continuity of parameter integrals)**.** *Let $G : T \to C([a, b])$ be continuous at t_0. Define*

$$F(t) = \int_a^b G(t)(x)\,dx.$$

Then F is continuous at t_0.

Proof. Let $I : C([a, b]) \to \mathbb{K}$ be the integral functional. Then

$$F = I \circ G.$$

Since G is continuous at t_0 and I is continuous, the composition is continuous. Quantitatively,

$$|F(t) - F(t_0)| = |I(G(t) - G(t_0))| \le (b - a)\|G(t) - G(t_0)\|_\infty \to 0.$$

□

Corollary 13.1 (Continuous kernel on a compact rectangle)**.** *If $f : [a, b] \times T_0 \to \mathbb{K}$ is continuous and T_0 is compact metric, then*

$$F(t) = \int_a^b f(x, t)\,dx$$

is continuous on T_0.

Proof. The set $[a, b] \times T_0$ is compact. By Cantor theorem, f is uniformly continuous on it. Hence $t \to t_0$ implies

$$\sup_{x\in[a,b]} |f(x, t) - f(x, t_0)| \to 0.$$

Thus the map $G(t) = f(\cdot, t)$ is continuous into $C([a, b])$, and the previous theorem applies. □

Definition 13.2 (Differentiability into a function space)**.** *Let $J \subset \mathbb{R}$. A map $G : J \to C([a, b])$ is differentiable at $t \in J$ with derivative $H \in C([a, b])$ if*

$$\left\| \frac{G(t + h) - G(t)}{h} - H \right\|_\infty \to 0 \quad (h \to 0).$$

Theorem 13.2 (Differentiation under the integral sign). *If $G : J \to C([a,b])$ is differentiable at t, then*

$$F(t) = \int_a^b G(t)(x)\,dx$$

is differentiable at t, and

$$F'(t) = \int_a^b G'(t)(x)\,dx.$$

Proof. Using linearity of I,

$$\frac{F(t+h)-F(t)}{h} - I(G'(t)) = I\left(\frac{G(t+h)-G(t)}{h} - G'(t)\right).$$

Taking absolute values and using boundedness of I, the absolute value is at most

$$(b-a)\left\|\frac{G(t+h)-G(t)}{h} - G'(t)\right\|_\infty \to 0.$$

□

Oral-exam minimum.

- Interpret $f(x,t)$ as a curve $G(t) = f(\cdot,t)$ in $C([a,b])$.
- Prove parameter continuity by composition with the integral functional.
- Prove differentiation under the integral sign by differentiability in norm.

Practice problems.

1. Let $G : T \to C([a,b])$ be continuous at t_0. Prove that $F(t) = \int_a^b G(t)(x)\,dx$ is continuous at t_0.
2. If $f : [a,b] \times T_0 \to \mathbb{K}$ is continuous and T_0 compact metric, prove that $t \mapsto f(\cdot,t)$ is continuous into $C([a,b])$.
3. Prove differentiation under the integral sign by treating $G : J \to C([a,b])$ as a differentiable curve in a normed space.
4. Work out an example where pointwise continuity in t is not enough to get continuity of $\int f(x,t)\,dx$ without some uniform control.

Ticket 14. Power series and local uniform convergence

Core idea. A power series has a radius inside which convergence is locally uniform. Local uniformity is what permits continuity, integration, and differentiation term by term.

Definition 14.1 (Power series). *A complex power series centered at z_0 is*

$$\sum_{n=0}^{\infty} a_n (z-z_0)^n.$$

Its radius of convergence is $R \in [0,\infty]$ such that the series converges absolutely for $|z-z_0| < R$ and diverges for $|z-z_0| > R$.

Theorem 14.1 (Cauchy–Hadamard). *For the power series* $\sum a_n(z-z_0)^n$,

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n},$$

with the conventions $1/0 = \infty$ *and* $1/\infty = 0$.

Proof. Let $L = \limsup |a_n|^{1/n}$. If $r < 1/L$, choose $q < 1$ with $rL < q$. For all large n, $|a_n|^{1/n} r < q$, hence $|a_n| r^n < q^n$. Therefore the series converges absolutely for $|z-z_0| \le r$.

If $r > 1/L$, then infinitely many n satisfy $|a_n|^{1/n} r > 1$, so $|a_n| r^n > 1$ infinitely often. The terms do not tend to zero, so the series diverges for $|z-z_0| = r$ along those radii. □

Theorem 14.2 (Local uniform convergence inside the disk). *For every* $0 < r < R$, *the series*

$$\sum a_n(z-z_0)^n$$

converges absolutely and uniformly on the closed disk $|z-z_0| \le r$.

Proof. Choose s with $r < s < R$. Since $s < R$, the numerical series $\sum |a_n| s^n$ converges. For $|z-z_0| \le r$,

$$|a_n(z-z_0)^n| \le |a_n| r^n \le |a_n| s^n.$$

The Weierstrass M-test gives absolute uniform convergence. □

Remark 14.1 (Filter language). For every compact disk $K_r = \{|z-z_0| \le r\} \subset \{|z-z_0| < R\}$, the partial sums form a Cauchy sequence in the Banach space $C(K_r)$ with the sup norm. Thus the power series is best understood as a locally uniform limit of polynomials.

Oral-exam minimum.

- State and prove Cauchy–Hadamard.
- Explain why convergence is locally uniform strictly inside the disk.
- Explain why no universal statement holds on the boundary $|z-z_0| = R$.

Practice problems.

1. Prove the Cauchy-Hadamard formula for the radius of convergence.
2. Prove that a power series converges locally uniformly inside its disk of convergence.
3. Give examples showing that no universal statement is possible on the boundary $|z-z_0| = R$.
4. Use the M-test to prove uniform convergence of $\sum a_n(z-z_0)^n$ on every closed disk $|z-z_0| \le r < R$.

Ticket 15. Termwise integration and differentiation of power series

Core idea. Inside every smaller closed disk, a power series is a uniform limit of polynomials with uniformly convergent derivative series.

Theorem 15.1 (Derivative radius is invariant). *The derivative series*

$$\sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}$$

has the same radius of convergence as

$$\sum_{n=0}^{\infty} a_n (z-z_0)^n.$$

Proof. By Cauchy–Hadamard, the reciprocal radius of the derivative series is

$$\limsup_{n\to\infty} |na_n|^{1/n} = \left(\lim_{n\to\infty} n^{1/n}\right) \limsup_{n\to\infty} |a_n|^{1/n} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Thus the radii agree. □

Theorem 15.2 (Termwise differentiation). *For $|z - z_0| < R$, the sum*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

is differentiable, and

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}.$$

Proof. Fix a closed disk $|z - z_0| \le r < R$. The original series and the derivative series converge uniformly on this disk by local uniform convergence and the invariant-radius theorem. Apply the termwise differentiation theorem for uniform limits on compact intervals in the real case, or the corresponding complex proof using difference quotients uniformly on smaller disks. Hence differentiation is termwise inside the convergence disk. □

Theorem 15.3 (Termwise integration). *For real x with $|x - x_0| < R$,*

$$\int_{x_0}^{x} \sum_{n=0}^{\infty} a_n (t - x_0)^n \, dt = \sum_{n=0}^{\infty} a_n \frac{(x - x_0)^{n+1}}{n+1}.$$

The integrated series has the same radius R.

Proof. On every closed interval $[x_0 - r, x_0 + r] \subset (x_0 - R, x_0 + R)$, the power series converges uniformly. Therefore the limit may be passed through the integral. The radius of the integrated series is unchanged because

$$(n + 1)^{1/n} \to 1.$$

□

Oral-exam minimum.

- Prove that the derivative series has the same radius.
- Prove termwise differentiation using uniform convergence of derivative series.
- Prove termwise integration using the integral functional.

Practice problems.

1. Prove that the derivative series $\sum_{n\ge1} n a_n (z - z_0)^{n-1}$ has the same radius of convergence as the original power series.
2. Prove termwise differentiation of a real power series on compact subintervals inside its interval of convergence.
3. Prove termwise integration of a power series and show that the integrated series has the same radius.
4. Apply termwise integration to derive the series for $\log(1 + x)$ from the geometric series for $1/(1 + x)$.

Ticket 16. Analytic functions, Taylor coefficients, and elementary expansions

Core idea. Analyticity means local equality with a power series. The coefficients are forced by derivatives, so power-series representations are unique.

Definition 16.1 (Analyticity). *A function f is analytic at x_0 if there exist $R > 0$ and coefficients (a_n) such that*

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \quad (|x - x_0| < R).$$

Theorem 16.1 (Uniqueness of power-series representation). *If*

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = \sum_{n=0}^{\infty} b_n (x - x_0)^n$$

for all x in a neighborhood of x_0, then $a_n = b_n$ for every n.

Proof. Subtract the two series. It is enough to prove: if

$$\sum_{n=0}^{\infty} c_n (x - x_0)^n = 0$$

near x_0, then all $c_n = 0$. Substituting $x = x_0$ gives $c_0 = 0$. For $x \neq x_0$, divide by $x - x_0$:

$$\sum_{n=1}^{\infty} c_n (x - x_0)^{n-1} = 0.$$

Letting $x \to x_0$, local uniform convergence justifies passing to the limit and gives $c_1 = 0$. Repeating inductively gives all coefficients zero. □

Corollary 16.1 (Taylor coefficients). *If*

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

near x_0, then

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

Proof. By termwise differentiation,

$$f^{(n)}(x) = \sum_{k=n}^{\infty} k(k-1)\cdots(k-n+1) a_k (x - x_0)^{k-n}.$$

Substitute $x = x_0$. Only the $k = n$ term remains:

$$f^{(n)}(x_0) = n! a_n.$$

□

Theorem 16.2 (Elementary series)**.** *For their usual radii of convergence,*

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!},$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \qquad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1),$$

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \quad (-1 < x \le 1).$$

Proof idea. The geometric series follows from finite geometric sums and passage to the limit. The logarithmic series follows by integrating the geometric series for $1/(1+x)$ term by term on compact subintervals of $(-1, 1)$, with endpoint $x = 1$ handled by the alternating series theorem or Abel-type limiting. The exponential, sine, and cosine series follow from solving the differential equations $f' = f$, $s' = c$, $c' = -s$ with the corresponding initial values and using uniqueness of power-series solutions. □

Theorem 16.3 (Euler formula)**.** *For every real x,*

$$e^{ix} = \cos x + i \sin x.$$

Proof. Using the absolutely convergent power series for e^z,

$$e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = \cos x + i \sin x.$$

□

Oral-exam minimum.

- Define analyticity.
- Prove uniqueness of power-series coefficients.
- Derive Taylor coefficients from termwise differentiation.
- Prove Euler's formula using absolute convergence.

Practice problems.

1. Prove uniqueness of power-series coefficients by induction using division by $x - x_0$.
2. Prove that if $f(x) = \sum a_n (x - x_0)^n$ near x_0, then $a_n = f^{(n)}(x_0)/n!$.
3. Derive the Taylor series for e^x, $\sin x$, and $\cos x$ using differential equations and initial values.
4. Prove Euler's formula $e^{ix} = \cos x + i \sin x$ from the power-series definitions.

Final checklist

Before the exam, make sure you can do the following without notes:

1. Define metric topology and neighborhood filters.
2. Prove closure through neighborhoods and explain the filter base $U \cap A$.

3. Define image filters and write continuity as $f_*\mathcal{N}_X(x) \to f(x)$.
4. Prove continuity preserves all convergent filters.
5. Explain why sequences characterize continuity only under first-countability.
6. Define metric uniformity and Cauchy filters.
7. Prove Cauchy-filter completeness equals sequential completeness in metric spaces.
8. Define ultrafilters and prove the ultrafilter compactness criterion.
9. Prove compact metric $\iff$ complete plus totally bounded.
10. Prove compact metric $\iff$ sequentially compact.
11. Write uniform continuity as $\mathcal{U}_Y \subset (f \times f)_*\mathcal{U}_X$.
12. Prove Heine–Cantor using an ultrafilter on $K \times K$.
13. Prove the uniform Cauchy criterion and Weierstrass M-test.
14. Prove uniform limits of continuous functions are continuous using filters.
15. Treat integration as a bounded linear functional on $C([a, b])$.
16. Treat parameter integrals as compositions $T \to C([a, b]) \to \mathbb{K}$.
17. Prove Cauchy–Hadamard and local uniform convergence of power series.
18. Prove termwise integration/differentiation and uniqueness of Taylor coefficients.

One-page dependency map

Concept	Structural meaning
Metric	Gives balls, topology, and metric uniformity.
Topology	Gives neighborhoods, closure, continuity, compactness, connectedness.
Neighborhood filter	Encodes "near a point"; $\mathcal{F} \to x \iff \mathcal{N}_X(x) \subset \mathcal{F}$.
Image filter	Encodes behavior of $f(x)$ along a filter: $f_*\mathcal{F}$.
Continuity	$f_*\mathcal{N}_X(x) \to f(x)$, equivalently all convergent filters are preserved.
Ultrafilter	Maximal filter; decides every subset.
Compactness	Every ultrafilter converges; equivalently closed families with FIP intersect.
Uniformity	Filter of neighborhoods of the diagonal in $X \times X$.
Uniform continuity	Pulls entourages back to entourages.
Cauchy filter	Eventually contained in sets of arbitrarily small diameter.
Completeness	Every Cauchy filter converges.
Uniform convergence	Convergence in a function-space uniformity, often the sup metric.

Integral theorem	Continuity of a bounded linear functional.
Parameter integral	Composition $T \to C([a,b]) \to \mathbb{K}$.
Power series	Locally uniform limits of polynomials.